

WEB-BASED VEHICLE AUCTION PLATFORM

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This dissertation is submitted in partial fulfilment of the requirement of the Degree of Bachelor of Information Technology (External) of the University of Colombo School of Computing

1.1 Declaration

I certify that this dissertation does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any University/Institute, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person or myself except where due reference is made in the text.

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1.2 Abstract

TaproJapan Co. Ltd., a Japan-based vehicle exporter, managed its vehicle auctions through WhatsApp broadcasts and informal messaging. This process offered no audit trail, no buyer verification, and frequent disputes over who placed the highest bid, creating a scaling bottleneck as the client's international buyer base grew. This project designed and implemented a web-based vehicle auction platform to replace this manual workflow with a structured, transparent, and auditable digital system.

The system was built as a three-tier full-stack application: a React 19 single-page front end, a layered Node.js/Express REST API, and a PostgreSQL relational database. The backend separates routes, controllers, services, and repositories, with all business logic such as authentication, role-based access control, bid validation, and auction status transitions concentrated in the service layer. The database schema evolved through eight incremental migrations, moving from a basic four-table design to a normalized structure supporting auction-scoped bidding, hidden reserve prices, and a three-state buyer verification workflow.

Key functionality delivered includes JWT-based stateless authentication, an admin-only buyer verification workflow, a vehicle catalogue with multi-criteria search and filtering, a time-bound auction module with dual-layer status correction, minimum-increment bid validation, reserve-price privacy protection for non-admin users, and an admin dashboard for vehicle, auction, and user management with manual or automatic winner selection. Direct-to-Cloudinary image uploads keep the backend stateless with respect to binary data.

The system was verified through unit and integration tests covering authentication, vehicle CRUD, bidding rules, and auction lifecycle transitions, supplemented by manual end-to-end and User Acceptance Testing with the client. Testing surfaced and resolved several non-trivial defects, including an authentication flow that blocked all logins, a role-escalation vulnerability in registration, and an architectural flaw that scoped bids to vehicles rather than auctions. The resulting platform meets its core functional and non-functional requirements and gives the client a scalable, transparent replacement for its manual auction process, with payment integration and shipping logistics identified as natural extensions for future work.

1.3 Acknowledgements

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1.6.1 List of Acronyms

API - Application Programming Interface

CRUD - Create, Read, Update, Delete

CDN - Content Delivery Network

CRC - Cyclic Redundancy Check (not used; placeholder removed)

ERD - Entity-Relationship Diagram

HTTP - Hypertext Transfer Protocol

JWT - JSON Web Token

ORM - Object-Relational Mapping

RBAC - Role-Based Access Control

REST - Representational State Transfer

SDLC - Software Development Life Cycle

SPA - Single Page Application

SQL - Structured Query Language

UAT - User Acceptance Testing

UI - User Interface

UUID - Universally Unique Identifier

Chapter 1 Introduction

1.1 Problem and Background

TaproJapan Co. Ltd. is a vehicle exporter based in Tokyo, Japan, operating as a small private business. The company purchases used vehicles at weekly Japanese auto auctions and resells them to international buyers in markets such as Sri Lanka, Kenya, and the Middle East. Prior to this project, the entire sales workflow of listing vehicles, collecting bids, and announcing winners was conducted manually through WhatsApp broadcast lists. Vehicle photographs and specifications were distributed informally with no consistent structure, buyers replied to message threads with their bids, and the client manually goes through these replies to determine the highest bidder. Winner selection was communicated privately, those buyers who did not win perceived as opaque and potentially unfair.

This manual process produced several serious problems: there was no audit trail of bids, so disputes about who bid what and when could not be resolved authoritatively; there was no access control, meaning any user could submit a bid without any verification of identity or seriousness; vehicle information was scattered across different message threads with no central record; there was no historical database of past auctions, vehicles, or buyer activity; and the process did not scale. As the number of vehicles and buyers grew, the owners' ability to track bids manually became an operational bottleneck.

1.2 Motivation

As TaproJapan decided to expand their buyer base internationally, they expected the increasing volume of concurrent bidding conversations to make manual tracking more error prone. Missed bids, duplicate entries, and miscommunication regarding vehicle specifications were likely to become more frequent, directly affecting customer trust and satisfaction. Furthermore, there was no mechanism to restrict bidding to verified and financially committed buyers, making it difficult for the client to distinguish serious purchasers from casual enquiries. This could slow down decision-making and increase the risk of a winning bidder failing to follow through with the purchase.

The client's intention to grow the business from a small operation into a larger international export operation required a professional, automated, and scalable platform that could attract and manage a larger pool of buyers without proportionally increasing manual administrative work. From an academic standpoint, the project offered an opportunity to model a genuine small-business process which is an auction-based commerce with identity verification and time-bound bidding, using a modern full-stack web architecture, and to confront real engineering trade-offs such as data consistency under concurrent bidding, stateless authentication, and reserve-price confidentiality.

1.3 Aim and Objectives

The aim of this project is to design, implement, and deliver a web-based vehicle auction platform that digitizes and automates TaproJapan's auction workflow, replacing informal messaging with a transparent, auditable, and scalable system.

The specific objectives are:

1. To develop a web-based platform that streamlines the client's vehicle auction process and improves transparency for buyers, by replacing ad-hoc messaging with a structured catalogue, auction, and bidding system.
2. To ensure secure and trustworthy participation by implementing proper user registration, admin-driven verification, and controlled, role-based bidding access.
3. To support efficient auction management by providing administrators with tools for handling vehicle listings, auction scheduling, user verification, and winning-bidder selection.
4. To maintain accurate historical records of vehicles, auctions, and bids, and to ensure data integrity, reliability, and secure handling of all stored information.
5. To validate and refine the system through iterative testing and direct stakeholder feedback so the delivered system aligns with the client's actual business needs.

1.4 Scope of the Project

The scope of this project covers the full development lifecycle of a web-based vehicle auction platform intended for use by TaproJapan Co. Ltd. and its international buyers. The system provides three tiers of access. Unauthenticated guests can browse the public vehicle catalogue and auction listings, search and filter vehicles, and view the complete public bid history for any auction. Registered buyers can additionally place bids, but only once their account has been verified by an administrator; buyers can also view auctions they have won on their profile page. Administrators are created only through a one-time seed script rather than through public registration and have full CRUD control over vehicle listings and auctions, can verify or reject buyer accounts, can monitor bidding activity in real time, and can close auctions and select winning bids either automatically or manually.

The system targets a small-to-medium volume of concurrent users appropriate to TaproJapan's current and near-term business scale, and is designed to be deployed as a single-server full-stack web application (Node.js/Express backend, PostgreSQL database, statically-served React front end behind Nginx). The environment assumed is a standard web browser on desktop or mobile, with no native mobile application component.

Explicitly out of scope for this project are payment gateway integration, shipping and logistics management, and advanced analytics or business-intelligence reporting beyond basic auction-result summaries. These were identified jointly with the client as valuable but separable concerns that can be added as a second phase once the core auction workflow is in active use, and are revisited in the Future Work section of Chapter 6.

1.5 Outline of the Dissertation

Chapter 2 analyses the existing manual process in detail, reviews comparable systems, and derives the functional and non-functional requirements, including security as a first-class functional requirement, along with the justification for the chosen development methodology and technology stack.

Chapter 3 presents the system design, covering the three-tier architecture, the layered backend pattern, the entity-relationship model and its evolution, and the user interface design.

Chapter 4 describes the implementation: the development environment, module structure, coding conventions, key algorithms such as auction status correction and reserve-price stripping, and version control practices.

Chapter 5 reports on the evaluation of the system, covering unit and integration testing, end-to-end manual testing, and user acceptance testing with the client, including the defects discovered and resolved during this process.

Chapter 6 concludes the dissertation with a critical evaluation of the project's outcomes, a personal reflection on the learning experience, and recommendations for future work.

Chapter 2 - Analysis

2.1 Existing System and Problem Description.

Before this project, TaproJapan's auction workflow ran entirely through informal communication channels, summarized in Table 2-1.

Table 2.1 Existing Manual Auction Workflow

Phase	**How it worked (old way)**
Vehicle Listing	Photos taken manually; specifications recorded on paper; distributed via WhatsApp broadcast lists.
Bid Collection	Buyers reply to message threads. The client manually reads all replies and identifies the highest bid.
Winner Selection	The client privately messages the winning bidder. The process is opaque to other participants.

As shown in Figure 2.1, the process begins with the acquisition and listing of a vehicle, followed by the distribution of vehicle details to potential buyers through informal communication channels. Buyers submit bids via message threads, which must be manually monitored by the client. Once the bidding period is considered complete, the client identifies the highest bidder and proceeds with the transaction offline. This workflow presents several limitations, including the absence of a defined bidding period, lack of structured bid records, limited transparency, and the inability to verify bidder authenticity. These issues reduce operational efficiency and make the process difficult to scale as the number of vehicles and buyers increases.

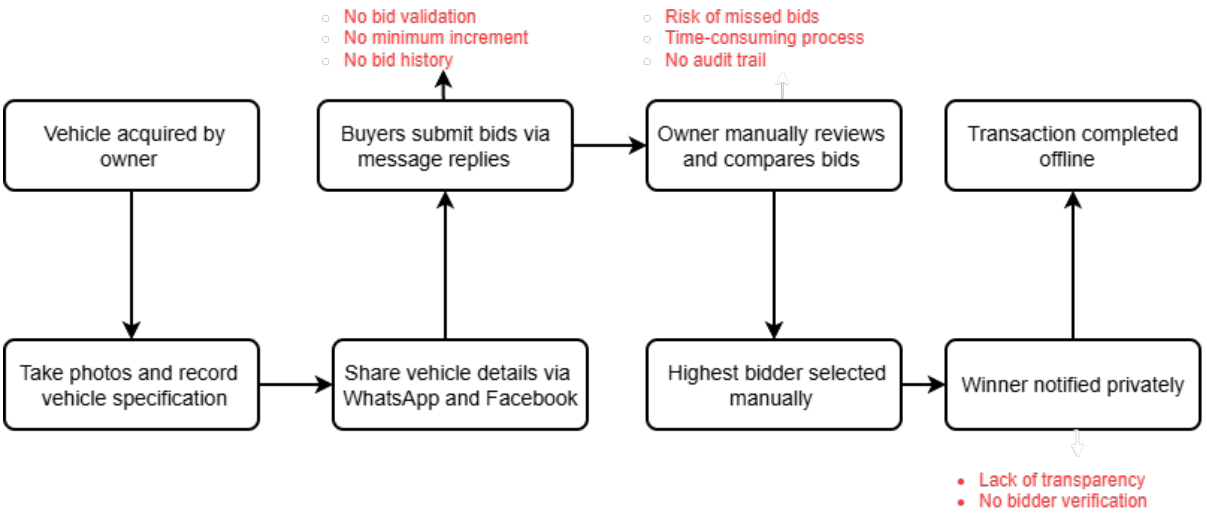


Figure 2.1 Existing Manual Vehicle Auction Workflow at TaproJapan

2.2 Review of Similar Systems

To inform the design of the platform, three categories of comparable system were reviewed: general consumer auction marketplaces, vertical vehicle-auction platforms, and verified-buyer marketplaces.

Consumer auction marketplaces (such as eBay-style time-bound auctions) provided the closest conceptual model for the core bidding mechanism. These systems typically employ fixed auction start and end times, competitive ascending bids, and publicly visible bid histories. This model was adopted for the proposed system because it promotes transparency and encourages competitive bidding among buyers.

Vertical vehicle-export auction platforms used by Japanese vehicle exporters typically incorporate vehicle-specific attributes such as chassis number, mileage, auction grade, and reserve prices. These platforms demonstrate the importance of structured vehicle data and auction-specific information management. Accordingly, the proposed system includes detailed vehicle records and support for reserve prices that remain hidden from non-admin users.

Verified-buyer marketplaces, commonly used in B2B and high-value asset trading environments, restrict participation to approved users rather than allowing open public bidding. This approach was particularly relevant to TaproJapan's requirements, as the

client wanted to ensure that only verified and financially committed buyers could participate in auctions.

In addition to reviewing these categories of systems, several existing vehicle auction platforms were examined to identify practical design and workflow patterns. Platforms such as Cars & Bids [11], AutocomJapan [12] , and RamaDBK[13] provide reference implementations of web-based vehicle auction platforms. Analysis of these systems reveals common design patterns: paginated vehicle grids with rich filtering, countdown timers displayed to buyers, bid ladders showing historical offers, and admin dashboards for lot management. These patterns have been incorporated into the UI/UX design of this project. Table 2.2 summarizes the comparison.

Table 2.2 Feature comparison of reviewed system categories

Feature	**Consumer auctions**	**Vehicle-export platforms**	**Verified-buyer marketplaces**	**This project**
Time-bound bidding	Yes	Yes	Sometimes	Yes
Hidden reserve price	Rare	Yes	Sometimes	Yes
Mandatory buyer verification	No	Sometimes	Yes	Yes
Public bid history	Yes	Sometimes	Rare	Yes
Minimum bid increment	Yes	Yes	Sometimes	Yes

The platform designed for this project therefore combines the open, transparent bid-history convention of consumer auctions with the verified-participation and hidden-reserve conventions of vehicle-export and B2B marketplaces, reflecting Tapro Japan's specific requirement for transparency toward genuine buyers without exposing commercially sensitive reserve information.

2.3 System Requirements Analysis

2.3.1 Functional Requirements

Functional requirements were elicited from direct discussion with the client and from the pain points identified in section 2.1. They are listed in Table 2.3 with priority and category.

Table 2.3 Functional requirements

ID	Requirement	Priority	Category
FR-01	User registration with email, password, and contact information	High	User Management
FR-02	Admin verification or rejection of registered user accounts	High	User Management
FR-03	Secure login with JWT-based authentication	High	User Management
FR-04	Role-based access control: public, verified buyer, admin	High	User Management
FR-05	Admin CRUD operations for vehicle listings (images, specs, auction date)	High	Vehicle Management
FR-06	Batch upload of weekly auction vehicles	Medium	Vehicle Management
FR-07	Public vehicle browsing with search and filter (make, model, year, price)	High	Vehicle Management
FR-08	Creation of auctions with configurable start/end datetime	High	Auction Management
FR-09	Automatic auction status transitions	High	Auction Management
FR-10	Verified users may place bids on active auctions	High	Bidding System

FR-11	Bid validation: minimum increment enforcement, auction timing check	High	Bidding System
FR-12	Real-time display of current highest bid to buyers	High	Bidding System
FR-13	Prevention of bidding after auction closure	High	Bidding System
FR-14	Complete bid history per auction, accessible to admin	High	Bidding System
FR-15	Admin dashboard: pending verifications, active auctions, bid summaries	High	Admin
FR-16	Admin selection of winning bidder per auction	High	Admin
FR-17	Email/in-app notifications for verification status and outbid events	Medium	Notifications
FR-18	Basic reporting: auction results, top buyers, bid counts (Appendix C - Management Report)	Medium	Reporting
FR-19	Self-service password reset via emailed, expiring, one-time token	Medium	User Management

2.3.2 Security as a Functional Requirement

Security was treated as a core functional requirement rather than a non-functional afterthought, since the system's central value proposition is trustworthy, auditable bidding. Four security controls were built directly into the system's functional behaviour. Access control is enforced through JWT-based role checks at the middleware layer (authRequired and requireRole), so every protected endpoint explicitly declares which roles may call it **[2]**. Admin account creation is excluded entirely from the public API surface; administrators can only be created by a database seed script run by a developer with direct database access, preventing any client-side request from escalating itself to admin. Input validation is enforced through Joi schemas at the route layer for every mutating endpoint, rejecting malformed or unexpected fields before they reach business logic **[7]**. Data protection is enforced through bcrypt password hashing (cost factor ≥ 10), following recommended password

storage practices **[3]**, parameterised SQL queries on every repository call to prevent injection, and reserve-price values being stripped from API responses at the service layer before they ever reach a non-admin client, rather than being hidden only in the front end. Treating these as functional requirements meant each had an explicit, testable behaviour rather than being an implicit assumption.

2.3.3 Non-Functional Requirements

Non-functional requirements specify the quality attributes and operational constraints of the system beyond its core functionality. These requirements ensure that the platform performs reliably, remains secure, and provides a satisfactory user experience. The non-functional requirements identified for the proposed system are summarised in Table 2.4.

Table 2.4 Non-functional requirements

Category	**Requirement**
Performance	API responses should complete in under 500 ms under normal load; the system should support at least 100 concurrent users.
Reliability	The system should maintain 99% uptime during active auction periods; database transactions must guarantee consistency for concurrent bid submissions.
Usability	The interface must be responsive across desktop and mobile screen sizes; returning buyers should require no training to use core features.
Scalability	The backend must be stateless (no server-side session store) so it can be deployed behind a load balancer if buyer volume grows; the database must use connection pooling.
Maintainability	The backend must enforce strict separation between routes, controllers, services, and repositories so that a change to one concern (e.g., a new database column) does not require changes to unrelated layers.
Security	See Section 2.3.2

2.4 Justification of Development Approach and Technologies

2.4.1 Chosen SDLC Model and Rationale

An iterative Agile-inspired development approach was adopted for this project. Although the work followed a broadly sequential progression from requirements analysis to deployment, development was carried out through multiple implementation and testing cycles, allowing features to be refined based on client feedback and system testing results.

The project commenced with requirements analysis and system design activities, including the preparation of the ER diagram, class diagram, sequence diagrams, and Software Architecture Design (SAD) documentation. Following completion of the design phase, backend services and database functionality were implemented, including user management, authentication, vehicle management, auction handling, and bid processing features.

Once the core backend functionality was operational, frontend development was carried out incrementally, integrating completed API endpoints into the user interface. The auction and bidding module received particular attention due to its importance to the overall system and the need to ensure correct bid validation, auction timing, and winner selection logic.

Testing was performed continuously throughout development, with dedicated efforts focused on unit testing, integration testing, security improvements, bug fixing, and user interface refinements. Several enhancements requested by the client were incorporated during later iterations.

This methodology was selected because the project involved a real client whose requirements evolved during development. The ability to implement, test, and refine features incrementally reduced project risk, enabled regular validation of system behaviour, and ensured that the final product aligned closely with the client's operational requirements. The major development phases are summarised in Table 2.5.

Table 2.5 Major Development Phases

Phase	**Deliverable**
Phase 1	Requirements Analysis and Problem Definition
Phase 2	System Analysis and Design (ERD, UML Diagrams, SAD)
Phase 3	Backend API and Database Development
Phase 4	Frontend Development and System Integration
Phase 5	Auction and Bidding Module Implementation
Phase 6	Testing, Security Improvements, and Bug Fixing
Phase 7	Documentation, Client Feedback Integration, and Final Deployment

2.4.2 Technology stack and justification

Table 2.6 Technology stack and justification

Layer	**Technology**	**Justification**
Frontend	React 19 + Vite + Tailwind CSS v4	A component-based SPA model suits a UI with many dynamic, frequently-updating regions (live bid amounts, countdown timers, status badges). Vite was chosen for its fast hot-module-reload during development. React was preferred over Vue for its larger ecosystem and longer-term maintainability for a single-developer project.[5]
Backend	Node.js 20 + Express 4	Non-blocking I/O is well suited to handling many concurrent bid submissions without thread-per-request overhead. Using JavaScript across both frontend and backend reduced context-switching for a solo developer. Express's minimalism allowed the strict custom layered architecture (Section 3.2) to be enforced explicitly rather than fighting a more opinionated framework. Django/Python was considered and rejected due to added deployment complexity without a corresponding benefit for this project's scale. [6]
Database	PostgreSQL 16	ACID-compliant transactions are essential to guarantee bid integrity under concurrent submissions. PostgreSQL's native array type (TEXT[]) was used directly for vehicle image URLs, avoiding a separate join table. Its NUMERIC type guarantees exact decimal arithmetic for currency, and its UNIQUE constraint semantics (treating each NULL as distinct) suited the optional chassis_number field. MySQL was considered and rejected due to weaker SERIALIZABLE isolation guarantees and the absence of a native array type.[4]

Authentication	JWT (jsonwebtoken)	A stateless token-based scheme avoids a server-side session store, simplifying horizontal scalability, and allows role and verification claims to travel with the request without an extra database lookup on every call [2].
Image storage	Cloudinary (free tier)	Unsigned browser-to-CDN upload keeps the backend stateless with respect to binary data and avoids the performance and backup complications of storing images in PostgreSQL or on the application server's local disk [9].
Validation	Joi	Declarative schema validation with built-in support for stripping unknown fields and collecting all validation errors in a single response, rather than failing on the first error [7].
Scheduling	node-cron	A lightweight in-process scheduler was sufficient at this project's scale; an external job queue was judged to be unnecessary complexity [8].

Chapter 3 - Design

3.1 Introduction to System Design

This chapter translates the requirements defined in Chapter 2 into a concrete architectural and data design. The guiding design principle throughout is separation of concerns: each layer of the backend, and each major frontend module, has exactly one responsibility, which directly supports the maintainability and testability non-functional requirements (Table 2.4).

3.2 Design Principles and Methodologies

The design of the vehicle auction platform was guided by several software engineering principles intended to improve maintainability, scalability, security, and reliability. Rather than focusing solely on implementing functional requirements, the system was structured to ensure that future modifications could be made with minimal impact on existing components. The principal design principles adopted are described below.

3.2.1 Separation of Concerns

The backend follows a strict separation of concerns by dividing responsibilities into four distinct layers: routes, controllers, services, and repositories. Each layer performs a single well-defined responsibility. Routes define HTTP endpoints and middleware chains, controllers handle request and response processing, services implement business rules, and repositories perform database operations using parameterised SQL queries. This separation reduces coupling between components, simplifies debugging, and enables individual layers to be tested independently without affecting the remainder of the application.

3.2.2 Layered Architecture

The application adopts a layered architecture consisting of presentation, application, and data tiers. Within the application tier, responsibilities are further divided into the route, controller, service, and repository layers. Client requests pass sequentially through these layers before reaching the PostgreSQL database. This organisation

isolates business logic from data access and user interface concerns, improving maintainability and allowing implementation details within one layer to change without requiring modifications to others.

3.2.3 RESTful and Stateless Design

Communication between the React frontend and the backend is implemented using RESTful web services over HTTP. Each request contains all information required to process it, including the JSON Web Token (JWT) used for authentication and authorisation. Because no session state is maintained on the server, the application remains stateless, simplifying deployment and allowing additional backend instances to be introduced without requiring shared session storage.

An alternative considered during design was the use of WebSockets (via Socket.io) to push bid updates to connected clients in real time, eliminating client-side polling entirely. This was evaluated against the system's stateless design principle: WebSockets require the server to maintain a persistent, addressable connection per client, which reintroduces server-side connection state and complicates horizontal scaling behind a load balancer; directly conflicting with the scalability non-functional requirement (Table 2.4). Given the project's expected concurrency (Section 2.3.3) and the client's small-to-medium buyer base, short-interval client polling (15 seconds) on the auction detail page was judged the more proportionate choice, preserving full backend statelessness at an acceptable cost to update latency. WebSockets remain a viable upgrade path if buyer volume grows to a scale where polling latency becomes user-facing (see Section 6.3).

3.2.4 Security by Design

Security considerations were incorporated throughout the system architecture rather than being added after implementation. Role-based access control restricts access to administrative functions, input validation is performed using Joi schemas before business logic executes, passwords are securely stored using bcrypt hashing, and all database operations use parameterised SQL queries to prevent SQL injection attacks. Sensitive information, such as auction reserve prices, is filtered within the service layer

before API responses are returned to non-administrative users, ensuring that confidential information cannot be exposed through client-side manipulation.

3.2.5 Modularity and Maintainability

The system was designed as a collection of independent modules corresponding to the major business entities: Users, Vehicles, Auctions, and Bids. Each module contains its own controller, service, repository, and route definitions, allowing features to be modified or extended without affecting unrelated functionality. This modular organisation also simplifies testing, code review, and future enhancement by reducing dependencies between different parts of the application.

3.2.6 Database Integrity

The database schema was designed to preserve data integrity through the use of primary keys, foreign keys, CHECK constraints, UNIQUE constraints, and ACID-compliant transactions. Relationships between users, vehicles, auctions, and bids are enforced at the database level to prevent orphan records and maintain referential integrity. Exact decimal monetary values are stored using PostgreSQL's `NUMERIC` data type to eliminate floating-point precision errors, ensuring reliable financial calculations during the bidding process.

3.3 System Architecture Overview

The system follows a three-tier architecture, illustrated conceptually in Figure 3.1 and detailed in Table 3.1.

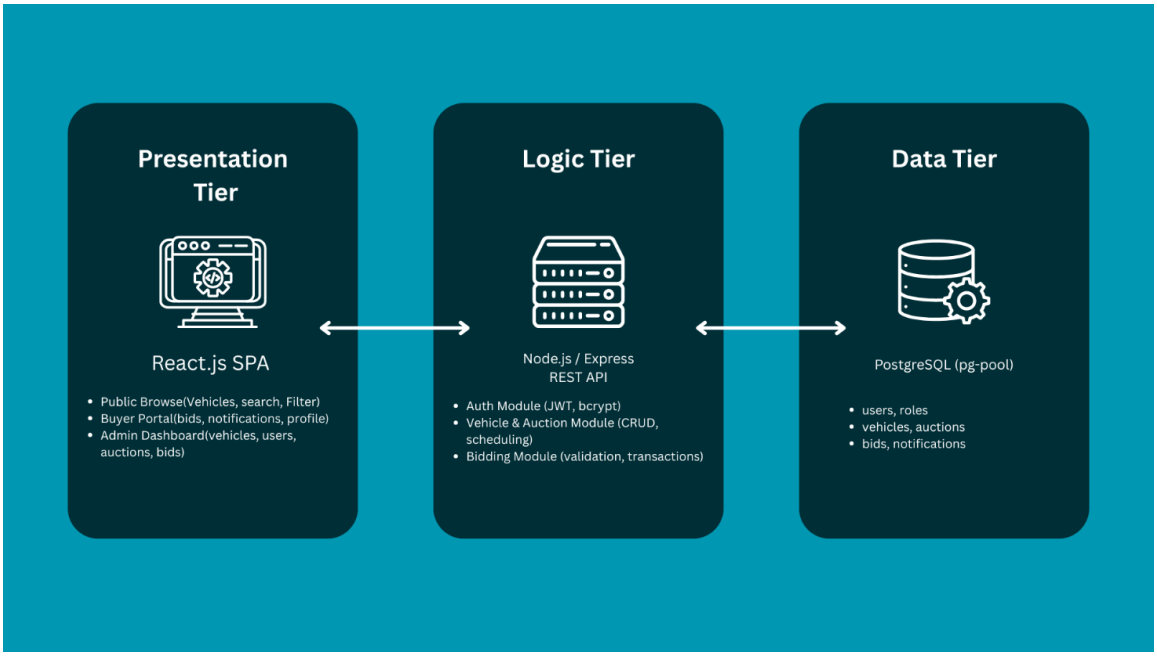


Figure 3.1 System High Level Architecture

Table 3.1 Three-tier architecture responsibilities

Tier	**Technology**	**Responsibility**
Presentation	React 19 + Vite + Tailwind CSS v4	Renders the UI, manages local component state, and communicates with the backend exclusively over HTTP via a single Axios instance. Contains no business logic.
Logic	Node.js + Express	Exposes a REST API [1]; enforces authentication, authorisation, and all business rules.
Data	PostgreSQL	Relational storage with ACID transactions for bid integrity, accessed through a pooled connection (pg.Pool).

Within the application tier, the backend follows a strict four-layer pattern, shown as a request trace in Figure 3.2

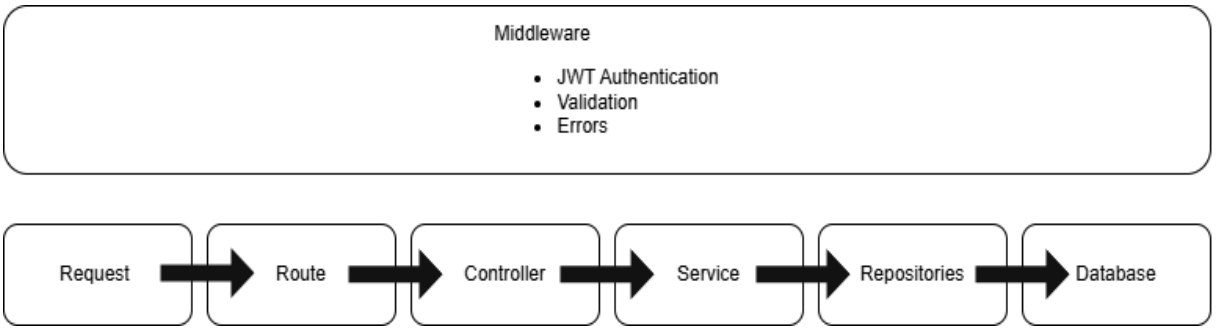


Figure 3.2 Backend Request Processing Architecture

Table 3.2 defines the responsibility of each layer precisely, since this separation is the central architectural decision of the backend.

Table 3.2 Backend layer responsibilities

Layer	**Location**	**Responsibility**
Routes	src/routes/	URL patterns and middleware chains only. Zero business logic.
Controllers	src/controllers/	Thin glue: call exactly one service method, then send the response or forward the error via next(err).
Services	src/services/	All business logic, authorisation decisions, and cross-entity rules.
Repositories	src/repositories/	SQL queries only, using parameterised statements. No business logic.
Middleware	src/middleware/	JWT verification, role enforcement, Joi validation, and global error handling.

This separation was chosen, rather than a more conventional MVC structure with logic embedded in controllers, for three reasons. First, testability: services can be unit-tested without spinning up the HTTP layer, and repositories can be mocked independently. Second, security: because all authorisation logic lives in the service layer rather than the controller, it is enforced consistently even if a service method is called from a different entry point (for example, from an automated test or, in future, from a background job), rather than relying on every controller to remember to re-check permissions. Third, change isolation: a schema change such as adding a new vehicle

column requires editing only the corresponding repository function; no other layer needs to know about column names.

3.4 Component and Module Design

3.4.1 Backend Modules

The backend is organised into one repository, one service, one controller, and one route file per primary resource: users, vehicles, auctions, and bids. Each repository exposes narrow, single-purpose functions (for example, findByEmail, findById, create, findAll, setStatus for the user repository). The auction repository's findAll and findById queries join the vehicles table to surface denormalised display fields (vehicle_make, vehicle_model, vehicle_year, vehicle_images) and use a correlated subquery, (SELECT MAX(amount) FROM bids WHERE auction_id = a.id) AS highest_bid, to compute the current highest bid live rather than maintaining a separately stored column that could fall out of sync with the bids table. The vehicle repository's findAll function supports dynamic filtering across up to nine optional conditions (status, make, model, year range, price range, free-text search), building a parameterised WHERE clause and running the query twice: once with COUNT(*) for pagination metadata and once with LIMIT/OFFSET for the page of results.

3.4.2 Frontend Modules

The frontend follows a similar single-responsibility convention. One Axios instance (src/api/client.js) carries a request interceptor that attaches the JWT Bearer token from localStorage to every outgoing request; one thin wrapper file per resource (auth.js, users.js, vehicles.js, auctions.js, bids.js, cloudinary.js) exposes typed functions over that instance, so no component is permitted to call Axios directly. Pages own their own data-fetching and loading/error state using useState and useEffect. Cross-cutting concerns that do not fit a parent/child component relationship , specifically, notifying the Navbar's pending-user-count badge when AdminPage approves or rejects a buyer are handled with a custom DOM event (window.dispatchEvent(new CustomEvent('user-status-changed'))) rather than prop drilling or a global state library, which was judged proportionate given the small number of cross-component signals required.

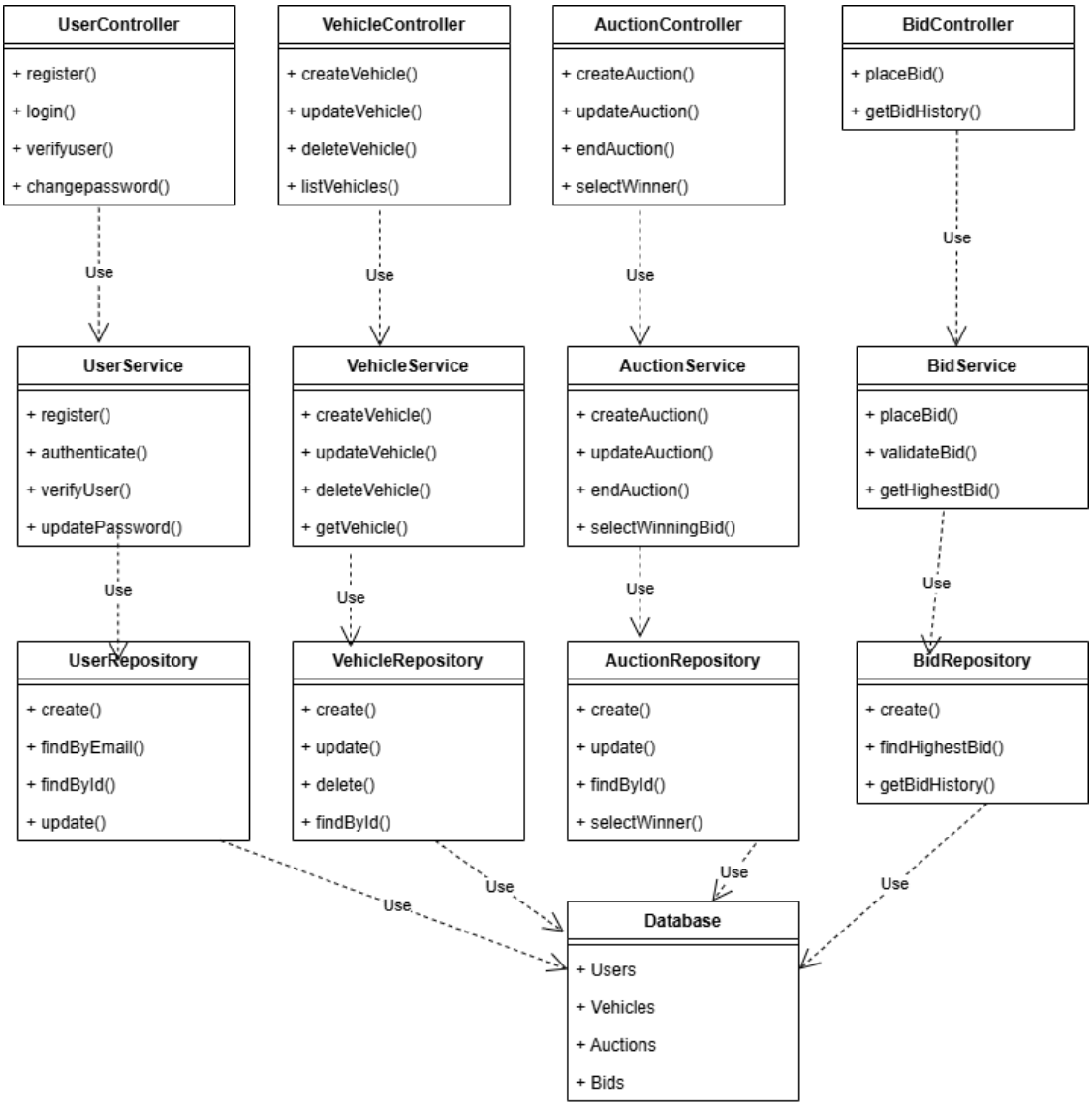


Figure 3.3 UML Class Diagram of the Layered Backend Architecture

Figure 3.3 illustrates the logical organisation of the backend components. Controllers receive HTTP requests and delegate processing to service classes, which implement business logic and interact with repository classes responsible for database operations. This layered organisation promotes modularity, maintainability, and easier testing by clearly separating responsibilities.

3.5 Workflow and Behavioural Modelling

3.5.1 End-to-End Sale Workflow

The vehicle auction process involves interactions between administrators, buyers, and the system itself. Figure 3.3 illustrates the complete lifecycle of an auction, beginning with vehicle listing creation and ending with winner notification. The workflow captures both user-driven activities and automated system operations such as auction activation and closure.

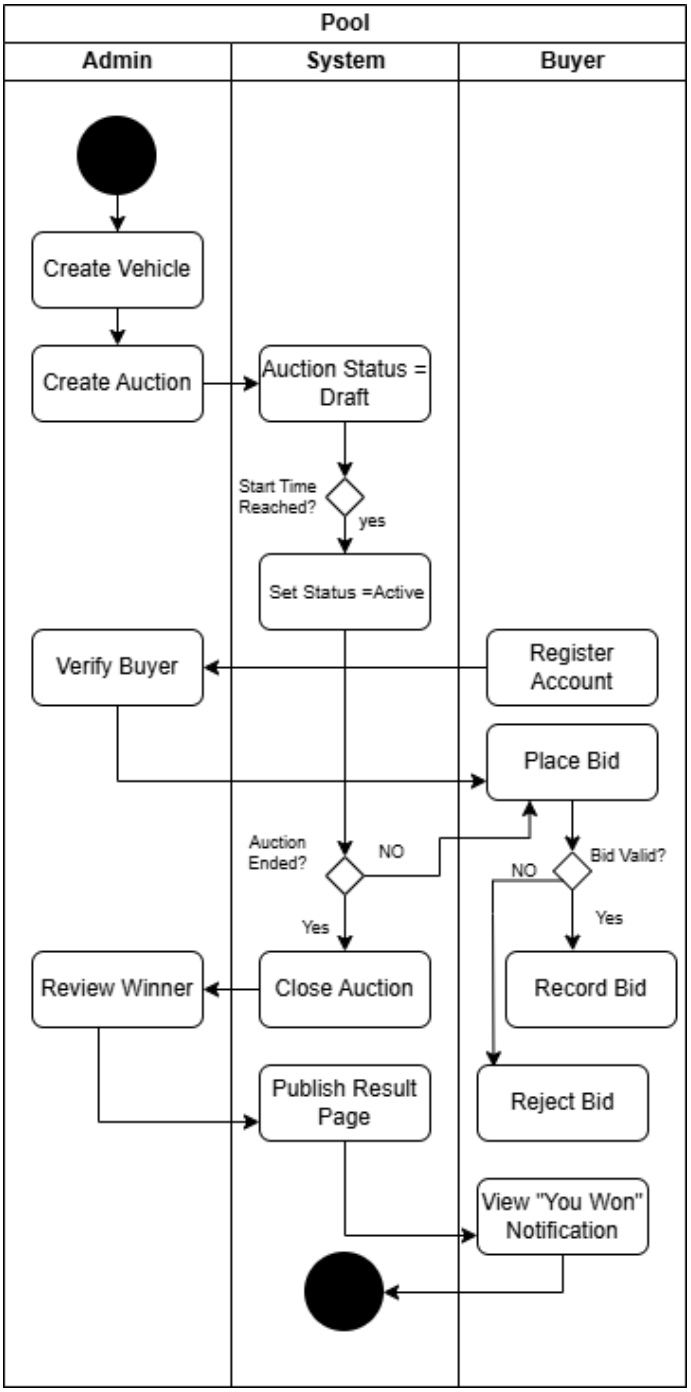


Figure 3.4 Activity Diagram for the End-to-End Vehicle Auction Workflow

As shown in Figure 3.3, the administrator first creates a vehicle listing and configures an associated auction by defining the auction period, reserve price, and minimum bid increment. The auction initially remains in the draft state until the configured start time is reached, at which point the system automatically activates the auction. Buyers must register and obtain administrator approval before they are permitted to participate in bidding.

During the active auction period, verified buyers may place bids that satisfy the validation rules enforced by the system. Bids that do not exceed the current highest bid by the required minimum increment are rejected. Once the auction end time is reached, the system closes the auction and identifies the highest bid. The administrator may then review the bid history and confirm or override the selected winner if necessary. Following winner confirmation, the auction result becomes publicly available through the winner page, and the successful bidder is notified through the application

3.5.2 Auction Status Transition Logic

Auction status is modelled as a three-state machine (draft, active, ended) driven by two independent mechanisms working together, since relying on either alone was found insufficient (see Bug #7 in Chapter 5). A cron job (node-cron, running every minute) executes two bulk UPDATE statements against the whole auctions table, transitioning draft auctions whose starts_at has passed to active, and active auctions whose ends_at has passed to ended. Independently, a pure function, effectiveStatus(auction), is evaluated on every read of an auction and recomputes the status from the stored timestamps; if it disagrees with the persisted status, the service immediately writes the corrected value back to the database. This guarantees that the status displayed to any client is always accurate at the moment of the request, while the cron job ensures the database itself self-corrects even when no request is being made.

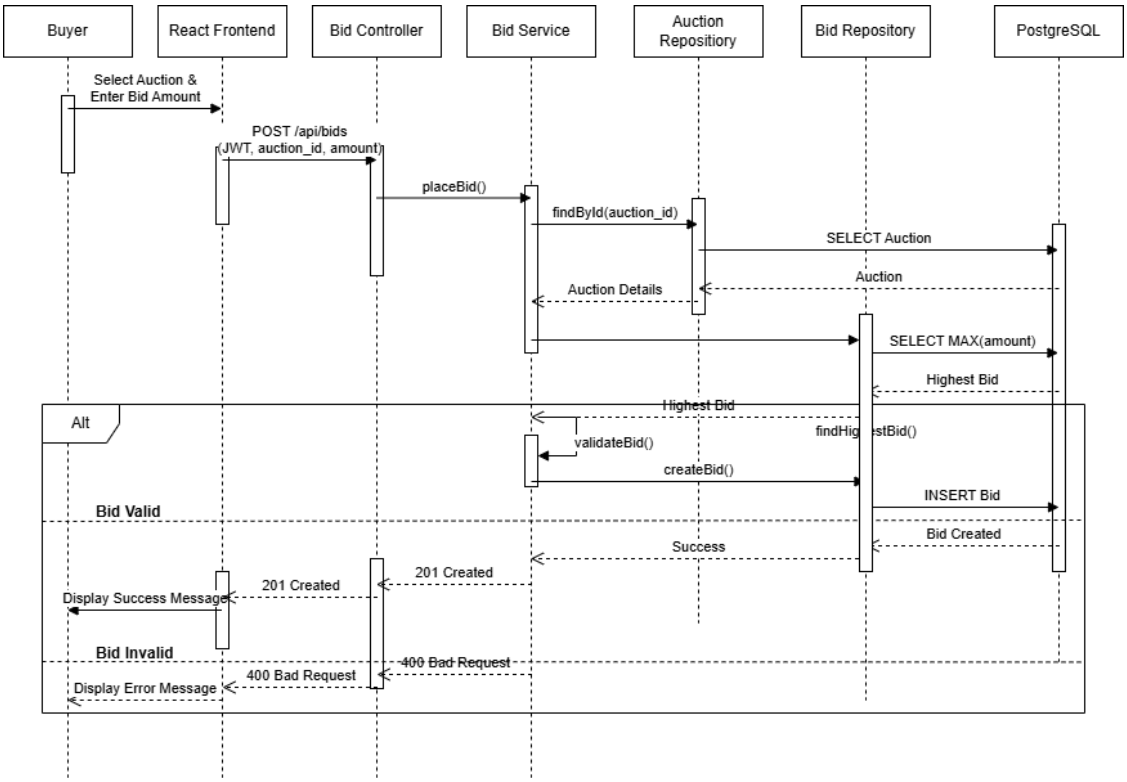


Figure 3.5 Sequence Diagram for Bid Placement

Figure 3.5 illustrates the interaction between the frontend, backend layers, and database during bid placement. The request passes through the controller, service, and repository layers before the bid is validated and persisted in the database. Alternative execution paths handle invalid bids and return appropriate error responses.

3.5.3 Key Data Modelling Decisions

Password reset tokens are stored as SHA-256 hashes rather than plaintext values. Since only the hash is persisted, an attacker gaining database access cannot use stored reset tokens to compromise user accounts. The original token exists only within the email sent to the legitimate account owner. Each token is associated with an expiry timestamp and is marked as used immediately after a successful password reset. This design enforces one-time usage, limits the exposure window of compromised links, and follows the same security principle applied to password hashing within the authentication subsystem.

3.6 Data Modelling

3.6.1 Entity-Relationship Overview

The initial database design was intentionally minimal and focused on validating the core auction workflow. As implementation progressed and additional requirements emerged, several limitations of the original schema became apparent. The original design stored bids directly against vehicles, which proved insufficient because bidding activity is associated with a specific auction event rather than the vehicle itself. This was corrected by introducing the auction_id foreign key on the Bid entity. Similarly, the original boolean is_verified attribute could only represent two states and was unable to distinguish between newly registered users awaiting approval and users who had been explicitly rejected. This limitation led to the introduction of the verification_status attribute with pending, verified, and rejected states.

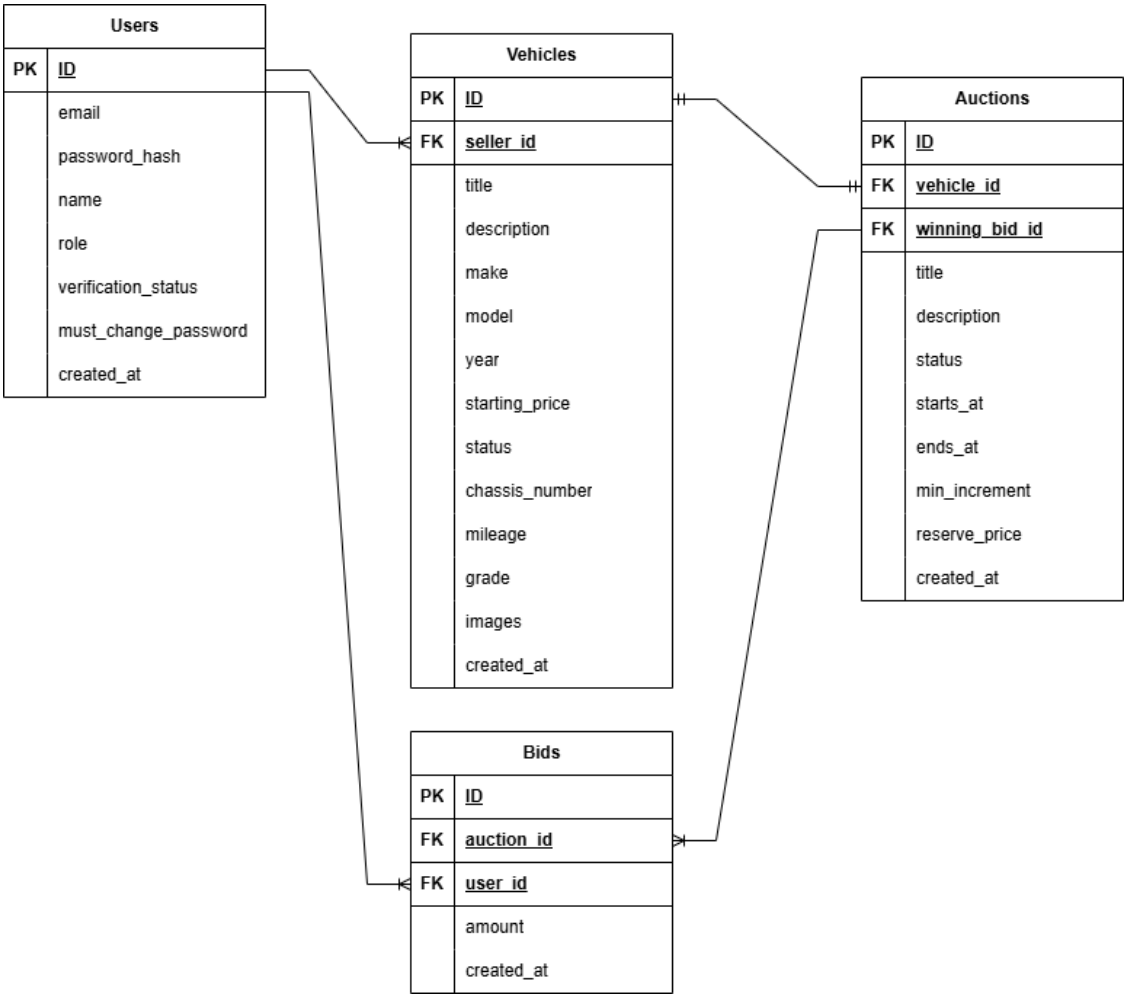


Figure 3.6 Entity-Relationship Diagram of the Database

Additional schema changes were driven by functional requirements identified during development. Vehicle-specific attributes such as chassis number, mileage, grade, and image collections were introduced to support detailed vehicle listings. The Auction entity was extended with min_increment and reserve_price attributes to support bid validation and hidden reserve-price functionality. Finally, the winning_bid_id relationship was introduced to explicitly record the selected winner of an auction, improving traceability and simplifying result publication. These changes transformed the original proof-of-concept schema into a production-oriented data model that more accurately reflected the business processes of Tapro Japan.

The final data model comprises four core entities: Users, Vehicles, Auctions, and Bids. An administrator user may own multiple vehicle listings through the seller_id foreign key. Each vehicle may be associated with an auction through the vehicle_id foreign key, while each auction may contain multiple bids linked through auction_id. Every bid belongs to exactly one user via the user_id foreign key. To support winner selection, the Auction entity optionally references a winning bid through the nullable winning_bid_id foreign key. Referential integrity is maintained through foreign key constraints, with cascading deletes applied where appropriate.

3.6.2 Schema Evolution

The schema was not designed in a single pass; it evolved through eight incremental, numbered migrations, each solving a specific problem identified during analysis or development, summarised in Table 3.3. This incremental approach follows the standard professional practice of never editing a migration after it has been applied to a real database every change is captured as a new file, preserving a complete, reproducible history of how the schema reached its final state.

Table 3.3 Migration history

Migration	**Change**	**Rationale**
001	Initial schema: users, vehicles, auctions, bids	Establishes the four core entities and basic auction workflow
002	Replace bids.vehicle_id with bids.auction_id	Bids belong to a specific auction event rather than directly to a vehicle
003	Add winning_bid_id to auctions	Records the winning bid selected for an auction
004	Add chassis_number, mileage, grade, images[] to vehicles	Supports detailed vehicle specifications and image management
005	Replace is_verified with verification_status	Supports pending, verified, and rejected account states
006	Add min_increment to auctions	Enforces minimum bid increment validation
007	Add reserve_price to auctions	Supports hidden reserve-price functionality
008	Add UNIQUE constraint on chassis_number	Prevents duplicate vehicle registrations
009	Add must_change_password to users	Supports mandatory password changes for temporary or seeded accounts

3.6.3 Key Data Modelling Decisions

NUMERIC over FLOAT [4] for currency. All monetary fields (starting_price, amount, min_increment, reserve_price) use NUMERIC(12,2) rather than a floating-point type. Floating-point binary representation cannot exactly represent all decimal fractions, which is unacceptable for financial values where every cent matters; NUMERIC stores an exact decimal value.

Computed highest_bid rather than a stored column. An earlier design considered storing current_highest_bid as a column on the auction, updated whenever a bid was inserted. This was rejected because it introduces a denormalization risk: if the bid insert succeeds but a subsequent update to the cached column fails (or is skipped due to a bug), the displayed highest bid silently diverges from the true value in the bids table. Computing it live via a correlated subquery removes this entire class of bug at a modest, acceptable query cost.

Enumerated verification_status over a boolean. The original design used a boolean is_verified flag. This could not represent the required three-state workflow (pending, verified, rejected) specifically, it could not distinguish a buyer who has registered but not yet been reviewed from one who has been explicitly rejected. Migration 005 replaced the boolean with a VARCHAR column constrained by a CHECK clause to these three values.

Inline CHECK constraints over lookup tables. An earlier ERD modelled role and auction/verification status as separate lookup tables with their own primary keys, following conventional normalisation [10] practice. This was deliberately simplified to inline CHECK constraints, because these are small, fixed sets of values that will never be added to or removed from at runtime; a lookup table would only add unnecessary JOIN overhead to the authentication path and every auction status query, with no corresponding gain in data integrity.

Eliminating the seller role. The original requirements analysis considered a distinct "seller" role separate from "admin". This was removed during analysis because, in Tapro Japan's actual business model, the owner is always the seller it is a one-to-many

relationship between one business and many buyers, not a marketplace with multiple independent sellers. Retaining a separate seller role would have added permission-checking complexity with no corresponding business value. The seller_id column was kept on the vehicles table (it always refers to the admin who created the listing, useful for audit purposes) but the role itself was removed from registration, route guards, and service logic.

3.7 User Interface Design

The interface was designed around three guiding principles: transparency (any visitor, logged in or not, can see the full public bid history and current highest bid for any auction), minimal friction for returning buyers (no training should be required , status is communicated through colour-coded badges and plain-language banners rather than jargon), and responsiveness across desktop and mobile, implemented with Tailwind CSS v4's utility classes rather than custom per-component CSS.

Key screens designed include: a public vehicle catalogue page with URL-synchronised filters (so that filtering, pagination, and even back/forward browser navigation all preserve and restore the correct query state) and skeleton-loading placeholders while data is in flight; an auction detail page , the most visually complex screen combining a lightbox image gallery, a live countdown timer, a colour-coded reserve-price indicator (no reserve / not met / met, with the actual amount visible only to admins), a bid submission form shown only to verified, authenticated, non-admin users while the auction is active, and (for admins) a "Close Auction" control and a bid ladder for manual winner selection; and an admin dashboard organised into three tabs (Vehicles, Auctions, Users), with the Users tab providing per-user verify/reject actions that immediately update a pending-count badge in the navigation bar via the custom event mechanism described in Section 3.3.2.

The admin interface also includes a reporting view providing operational, auction-performance, buyer-activity, and vehicle-inventory summaries (Appendix C - Management Report), fulfilling FR-18.

Chapter 4 - Implementation

4.1 Development and Deployment Environment

Development was carried out on Microsoft Windows using Visual Studio Code as the primary development environment. The backend was implemented using Node.js 20 LTS, PostgreSQL 16, and ECMAScript Modules, with dependencies including Express, pg, bcrypt, jsonwebtoken, Joi, Morgan, node-cron, and dotenv. Nodemon was used during development to automatically restart the server when source files changed. The frontend was developed using React and Vite, with React Router for client-side routing, Axios for API communication, and Tailwind CSS for styling. Nodemailer was used to send password reset emails through the configured SMTP server. Git was used throughout the project for version control and change tracking.

Environment-specific configuration such as database connection strings, JWT secrets, and Cloudinary credentials was managed through environment variables stored in .env files and excluded from version control. This approach ensured that sensitive information was not exposed within the source repository.

The backend application and PostgreSQL database were deployed using Railway, which provides managed application hosting, automated builds, and database services. The frontend application was deployed separately on Vercel, which provides static hosting and content delivery network (CDN) support for React applications. This deployment architecture simplified infrastructure management by eliminating the need for manual server provisioning, reverse proxy configuration, and process management while still providing reliable application hosting and automated deployment from the Git repository.

4.2 Development Practices and Coding Standards

The codebase is organised according to the four-layer architecture described in Section 3.2, with each resource implemented through a dedicated route, controller, service, and repository. A consistent naming convention is followed throughout the project: repository, service, and controller files are named according to the resource they manage (e.g., `vehicleRepository.js`, `auctionService.js`, and `bidController.js`), while route files are named after the corresponding resource in plural form (e.g., `vehicles.js` and `auctions.js`).

Environment variables are accessed through a single configuration module, `src/config/index.js`, rather than being read directly via `process.env` throughout the application. This centralised approach improves maintainability and simplifies code review by ensuring that configuration values and secrets are managed consistently. The application entry points are separated into two files: `app.js`, which constructs the Express application and registers middleware and routes, and `server.js`, which starts the HTTP server and cron scheduler. This separation allows automated tests to import `app.js` directly without unintentionally starting background services or opening network ports.

Error handling follows a uniform convention across the service layer. When a business-rule violation is detected, the service throws a standard `Error` object with an attached `status` property (for example, `err.status = 404`). A single global error-handling middleware then reads this property and generates the corresponding HTTP response. This approach ensures that the meaning of each error is defined once within the layer that detects it, rather than being repeatedly interpreted by individual controllers, resulting in more consistent and maintainable error handling throughout the application.

4.3 Critical Code Segments

4.3.1 Authentication Middleware

Two exported middleware functions guard protected routes. `authRequired` reads the `Authorization: Bearer <token>` header, calls `jwt.verify() [2]`, and, on success, sets `req.user` to the decoded payload containing the user's database ID (`sub`), role, email, and verification status. On failure, it returns an HTTP 401 (Unauthorized) response. `requireRole(roles)`, which must execute after `authRequired`, checks whether `req.user.role` is included in the allowed roles array and returns an HTTP 403 (Forbidden) response otherwise. These middleware functions are composed declaratively within route definitions. For example:

```
router.post(

  '/',
  authRequired,
  requireRole(['admin']),
  validate(vehicleSchema),
  createVehicleController
);
```

This approach makes the access-control rules for each endpoint directly visible within the route definition.

4.3.2 Auction Status Correction

The `effectiveStatus()` function in `auctionService.js` is the core algorithm underpinning the dual-layer status-transition mechanism described in Section 3.4.2:

```
export function effectiveStatus(auction) {
  const now = new Date();
  if (
    auction.status === 'draft' &&
    auction.starts_at &&
    new Date(auction.starts_at) <= now
  ) {
    return 'active';
  }
```

```
  if (
    auction.status === 'active' &&
    auction.ends_at &&
    new Date(auction.ends_at) <= now
  ) {
    return 'ended';
  }

  return auction.status;
}
```

This function is pure and has no side effects, making it straightforward to unit test against a range of timestamp combinations. The calling service compares the computed status against the persisted database value and issues a write-back update only when the two values differ, thereby minimising unnecessary database writes.

4.3.3 Reserve Price Privacy

Reserve-price confidentiality (FR-15) is enforced at the service layer rather than relying on the frontend to hide sensitive information:

```
function stripReserve(auction, user) {
  const reserveMet =
    auction.reserve_price == null
      ? null
      : (
          auction.highest_bid != null
            ?              Number(auction.highest_bid)              >=
Number(auction.reserve_price)
            : false
        );

  if (user?.role === 'admin') {
    return { ...auction, reserve_met: reserveMet };
  }

  const { reserve_price, ...rest } = auction;
  return { ...rest, reserve_met: reserveMet };
}
```

Administrators receive the complete auction object, including the reserve price. All other users receive the same object with the `reserve_price` field removed, while still receiving a `reserve_met` indicator (or `null` if no reserve price exists). This allows buyers to view meaningful reserve-status information without exposing the underlying reserve amount.

4.3.4 Bid Placement Validation

`bidService.placeBid()` performs an ordered sequence of validation checks, each throwing a descriptive error with an appropriate `.status` value on failure. First, the user's verification status is checked using information stored in the JWT payload, avoiding an additional database query. Next, the existence of the referenced auction is verified. The service then evaluates `effectiveStatus(auction)` to ensure that the auction is currently active, distinguishing between auctions that have not yet started and those that have already ended so that accurate error messages can be returned. Finally, the submitted bid amount is validated to ensure that it exceeds the current highest bid by at least the configured `min_increment` value, or simply exceeds the highest bid when `min_increment` is zero.

This validation order,verification, existence, timing, and amount,was chosen so that the most common and informative failure condition, an unverified user attempting to bid, is reported before further validation logic is executed.

4.3.5 Partial Update Merging

`vehicleService.updateVehicle()` (and the equivalent auction-update implementation) first retrieves the existing database record and merges it with the incoming payload using the nullish-coalescing operator rather than passing the raw request body directly to an `UPDATE` statement:

```
const merged = {
  title: payload.title ?? existing.title,
  make: payload.make ?? existing.make,
  // remaining fields follow the same pattern
};
```

This approach uses the value supplied in the request only when it has been explicitly provided, falling back to the existing database value otherwise. The pattern supports partial updates through HTTP PUT/PATCH endpoints and was introduced specifically to resolve the defect described in Chapter 5 (Bug #6), where omitted fields were being unintentionally overwritten with `NULL`.

4.3.6 Password Reset Token Handling

The password recovery feature follows a secure token-based workflow. When a user submits a password reset request, the server generates a cryptographically secure random token and computes its SHA-256 hash before storing it in the database. Only the original token is sent to the user's registered email address using Nodemailer. During password reset, the received token is hashed again and compared against the stored hash. The system verifies that the token has not expired and has not been previously used before allowing the password update. After a successful reset, the token is immediately invalidated to prevent reuse. This implementation minimizes the impact of database compromise while providing a convenient self-service recovery mechanism for users.

4.4 Version Control Practices

Git was used throughout the project to manage source code, track changes, and support incremental development. The repository was organised into two primary directories, `Backend/` and `Frontend/`, each containing its own dependencies and configuration files. A root-level `.gitignore` file excluded generated dependencies (`node_modules/`), environment configuration files (`.env`), build artefacts (`dist/`), and operating-system-specific files from version control.Commits were created regularly throughout development, with commit messages describing the functionality, bug fixes, testing activities, security improvements, and client-requested changes being implemented. The commit history therefore provides a chronological record of the system's evolution from requirements analysis and system design through implementation, testing, deployment, and maintenance activities.

Database schema changes were managed through numbered migration files. Once a migration had been applied to a database instance, it was never modified directly.

Instead, all subsequent schema changes were introduced through new migration files, preserving a complete and reproducible history of database evolution as described in Section 3.5.2.

4.5 Testing Tooling Notes

Several environment-specific considerations were encountered during testing and were resolved to ensure reproducibility across development environments. The backend application was implemented using ECMAScript Modules (`"type": "module"`), which required additional configuration when running tests with Jest. Because Jest's default execution model is based on CommonJS modules, tests were executed using Node.js's `--experimental-vm-modules` flag to enable compatibility with ECMAScript Modules.

A second issue was encountered on Windows systems when invoking Jest through the executable located in `node_modules/.bin`. The generated launcher script is intended for POSIX-compatible shells and therefore produced execution errors when invoked directly in certain Windows environments. This issue was resolved by executing Jest through its JavaScript entry point located in `node_modules/jest/bin/jest.js`, providing a platform-independent solution.

The final test command used during development was:

```
node --experimental-vm-modules node_modules/jest/bin/jest.js --runInBand -forceExit
```

This configuration ensured consistent execution of automated tests across the development environment while maintaining compatibility with the project's ECMAScript Module architecture.

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Chapter 5 - Evaluation

5.1 Testing Strategy Overview

The system was verified at four levels: unit testing of pure business-logic functions, integration testing of complete API request/response cycles, manual system testing via a structured end-to-end checklist, and User Acceptance Testing with the client. A dedicated test database (vehicle_auction_test), schema-identical to the development database but populated independently, was used for all automated tests, with its own .env.test configuration, so that test runs could never affect or be affected by development data.

5.2 Unit and Integration Testing

Automated tests were written using Jest, organised one test file per resource, as summarised in

Table 5.1 Automated test coverage by file

Test file	**Coverage**
auth.test.js	Buyer registration, rejection of attempted admin self-registration, duplicate-email handling, password validation, login and token issuance, and 401 responses for missing/invalid credentials
vehicles.test.js	Public listing, and create/update/delete operations verified against 401/403/200/204 outcomes per role, plus the filtering and pagination query suite
users.test.js	Verify/reject workflow, the corresponding 401/403/404/409 error cases, confirmation that a freshly issued token correctly reflects isVerified: true after verification, and the /users/me endpoint
bids.test.js	Rejection of bids from unverified buyers, successful bids from verified buyers, the outbid/minimum-increment enforcement rule, and rejection of bids placed by an admin account
auctions.test.js	Auction CRUD, the close-auction action, manual winner selection, the public winner endpoint, and the automatic status-transition logic

Because services were designed to be callable independently of the HTTP layer (Section 3.2), pure functions such as effectiveStatus() could be tested directly against a range of timestamp inputs without needing to mock an HTTP request at all, which kept

these particular tests fast and deterministic.

```
PS C:\Personal Files\Project BIT\Project\backend> npm test

> vehicle-auction-backend@0.1.0 test
> node --experimental-vm-modules node_modules/jest/bin/jest.js --runInBand --forceExit

(node:12988) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
 PASS  src/__tests__/vehicles.test.js
 PASS  src/__tests__/auth.test.js
 PASS  src/__tests__/auctions.test.js
 PASS  src/__tests__/bids.test.js
 PASS  src/__tests__/users.test.js

Test Suites: 5 passed, 5 total
Tests:       105 passed, 105 total
Snapshots:   0 total
Time:        6.052 s
Ran all test suites.
Force exiting Jest: Have you considered using `--detectOpenHandles` to detect async operations that kept running after all
tests finished?
```

Figure 5.1 Successful completion of tests

5.3 System and End-to-End Testing

Backend endpoints were additionally exercised manually with ‘curl’ against a running development server, covering registration, login, admin user-verification, vehicle and auction creation, and bid placement, to confirm behaviour matched the automated test expectations under a real running process rather than only inside the test harness.

A structured frontend smoke-test checklist was run through after each significant feature was completed, covering the full lifecycle in sequence: registering a new buyer and confirming the redirect to login with a success message; logging in as admin and confirming redirection to the admin dashboard; creating a vehicle with at least two Cloudinary-hosted images; creating an active auction with a reserve price against that vehicle; confirming the bid form is hidden for an unverified buyer; verifying the buyer as admin and confirming the navbar's pending-count badge decrements immediately via the custom event mechanism; confirming the bid form becomes visible only after the buyer re-logs-in (refreshing their JWT); placing a bid and confirming it appears in the public bid list with the reserve indicator updating accordingly; closing the auction as admin and confirming the highest bid is automatically selected as winner; confirming the public winner page is reachable and correct; logging in as the winning buyer and confirming the "You Won!" banner appears; confirming the won auction appears on the profile page with the correct winning amount; and confirming that dismissing the win banner persists across a subsequent page load (via the ‘wonDismissed’ key in localStorage).

5.4 Performance Testing

To assess the system's responsiveness under concurrent load, a load-testing script was written and executed against the local development server. The script issued 100 concurrent requests to each of four representative endpoints ;the health check, filtered vehicle search, paginated vehicle inventory listing, and auction listing, with a 10-request warm-up phase per endpoint beforehand to exclude cold-start effects such as connection-pool initialisation. For each endpoint, the average, 95th-percentile (P95), and maximum response times were recorded. Table 5.2 summarises the results.

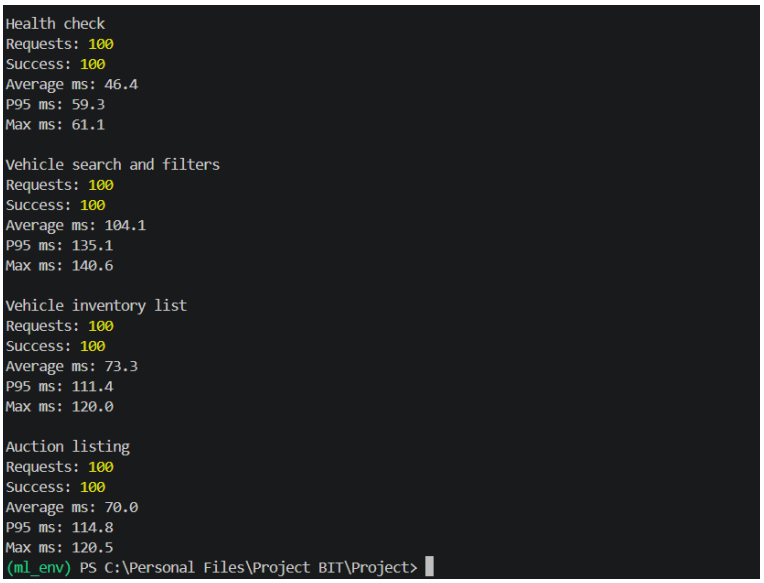


Figure 5.2 Performance Testing on local server

All four endpoints returned a 100% success rate across the 100 concurrent requests, and all response times remained well within the 500ms performance target defined in Table 5.2. The filtered vehicle search endpoint showed the highest latency of the four, consistent with it performing the most complex query - a dynamically constructed WHERE clause across up to nine optional filter conditions, executed twice (once for the result page, once for the pagination count).

These results were measured against the local development environment (Node.js/Express backend and PostgreSQL running locally) rather than the deployed Railway instance, and therefore do not account for real network latency, Railway's container resource limits, or production database connection-pooling behaviour. Testing against the production deployment was identified as a valuable extension of this work but was not completed within the project timeline; this is revisited in section 6.3.

Table 5.2 Local load-test results by endpoint

Endpoint	Requests	Success	Avg (ms)	P95 (ms)	Max (ms)
Health check	100	100	46.4	59.3	61.1
Vehicle search and filters	100	100	104.1	135.1	140.6
Vehicle inventory list	100	100	73.3	111.4	120.0
Auction listing	100	100	70.0	114.8	120.5

5.5 Defects Identified and Resolved

Systematic review of the implementation against the requirements in Chapter 2 surfaced twelve significant defects during development, several of which were architecturally important rather than cosmetic. These are recorded here because they represent the most substantive technical learning from the implementation phase.

The most severe defect found was a complete absence of any database schema , the migrations directory was initially empty while the rest of the application assumed the four core tables already existed, causing the server to crash on the very first API call. This was resolved by writing ‘001_initial_schema.sql’, with all subsequent schema changes made only through additively numbered migration files thereafter.

A second blocking defect was discovered in the authentication flow: ‘authService.login()’ contained a hard 403 rejection for any user whose ‘is_verified’ flag was false, but no endpoint existed anywhere in the system to ever set that flag to true, meaning every user who registered was permanently locked out of logging in at all. The intended behaviour, that unverified users can log in but cannot bid , had been implemented as a complete login block instead. This was fixed by removing the 403 check from login entirely, embedding the verification status in the JWT issued at login time, moving the verification check specifically into ‘bidService.placeBid()’ where it belonged, and adding the missing admin-only ‘PATCH /api/users/:id/status’ endpoint.

A security-significant defect allowed any unauthenticated client to self-register as an administrator, because the registration endpoint's Joi schema accepted ‘role: 'admin'‘ as a valid value submitted directly in the request body. This was closed with a two-layer fix: the route-level schema was restricted to accept only ‘'buyer'‘, and a second, independent guard was added inside ‘authService.register()’ checking submitted roles

against an explicit allow-list, so the rule holds even if route-level validation were ever bypassed or refactored incorrectly. This reflects the general security principle, applied consistently throughout the system, that a privileged role is a trust boundary decided by the server, never a value the client is trusted to supply.

A logic gap meant bids were being accepted against auctions in any state , including auctions that had not yet started, had already ended, or did not exist at all , because ‘bidService.placeBid()’ contained only a TODO comment where the auction-state check should have been. This was fixed by having ‘placeBid()’ fetch the auction and evaluate ‘effectiveStatus()’ before accepting any bid amount, with distinct error messages for the not-yet-started and already-ended cases.

An architectural defect, more serious than a simple bug, was that the original bids table scoped bids to a vehicle (‘vehicle_id’) rather than to a specific auction event (‘auction_id’). Because a single vehicle could in principle be re-auctioned, this meant a query for the "current highest bid" on a vehicle could incorrectly return a bid left over from a previous, unrelated auction of that same vehicle. This was corrected via migration 002, which dropped ‘vehicle_id’ from the bids table entirely and introduced ‘auction_id’ as the sole scoping foreign key, with supporting indexes added for the now auction-scoped highest-bid query.

A further logic defect caused partial vehicle updates to silently destroy data: ‘vehicleService.updateVehicle()’ originally passed the raw request body straight through to an UPDATE statement that set every column, so a client sending only ‘{ title: "New Title" }’ would have every other field make, model, description, images , overwritten to NULL. This was fixed with the existing-record merge pattern described in Section 4.3.5.

Auction status was found to never change automatically at all , the ‘starts_at’/’ends_at’ timestamps and ‘status’ column existed, but nothing in the system actually transitioned status based on the current time, so an auction could remain "draft" indefinitely even after its scheduled start had passed. This led to the dual-layer fix (cron scheduler plus on-read correction) described in Section 3.4.2 and Section 4.3.2.

No mechanism existed at all to record which bid had won an auction , there was no ‘winning_bid_id’ column and no service method to select one , meaning the central business requirement of the entire project, selecting and announcing a winner, was simply unimplemented. This was resolved with migration 003 and the addition of ‘closeAuction()’ (automatic highest-bid selection) and ‘selectWinner()’ (manual admin override) service methods, together with a public ‘GET /api/auctions/:id/winner’ endpoint.

There was no endpoint to retrieve a user's current, live profile data , only the (potentially stale) claims embedded in their JWT , which was insufficient for the profile page's verification-status display. ‘GET /api/users/me’ was added to close this gap.

The auction routes had no Joi input validation at all on create or update, in contrast to the vehicle routes, meaning arbitrary request bodies reached the service layer unchecked; explicit ‘auctionCreateSchema’ and ‘auctionUpdateSchema’ schemas were added to close this gap.

A subtler defect was found through the automated test suite rather than manual inspection. The authService.register() response constructed the API field isVerified using user.isVerified, but the underlying database row exposed the snake_case column is_verified. As a result, every registration response silently returned isVerified: undefined. This defect was detected by an explicit test assertion (expect(res.body.isVerified).toBe(false)) and is recorded here as evidence of the practical value of automated testing beyond manual smoke testing.

Finally, the bid-history endpoint (‘GET /api/auctions/:id/bids’) had been placed behind ‘authRequired’, which silently locked out guests and any logged-out visitor from seeing bid history at all , directly contradicting the transparency requirement (FR-12) that bid history be public. This was fixed by removing the authentication requirement from that specific route and confirming the frontend fetches bid history unconditionally rather than only when a user is logged in.

5.6 User Acceptance Testing

Following the resolution of the defects above, a User Acceptance Testing session was conducted with the client. The two business owners at Tapro Japan Co. Ltd. were given guided access to a deployed instance of the system and asked to perform representative tasks corresponding to their actual weekly workflow: listing a newly acquired vehicle with photographs and specifications, scheduling an auction for it, reviewing the live bid ladder for an active auction, verifying a newly registered buyer, and closing an auction to select a winner. Feedback was gathered through direct observation and follow-up discussion rather than a formal written survey, given the small number of stakeholders involved.

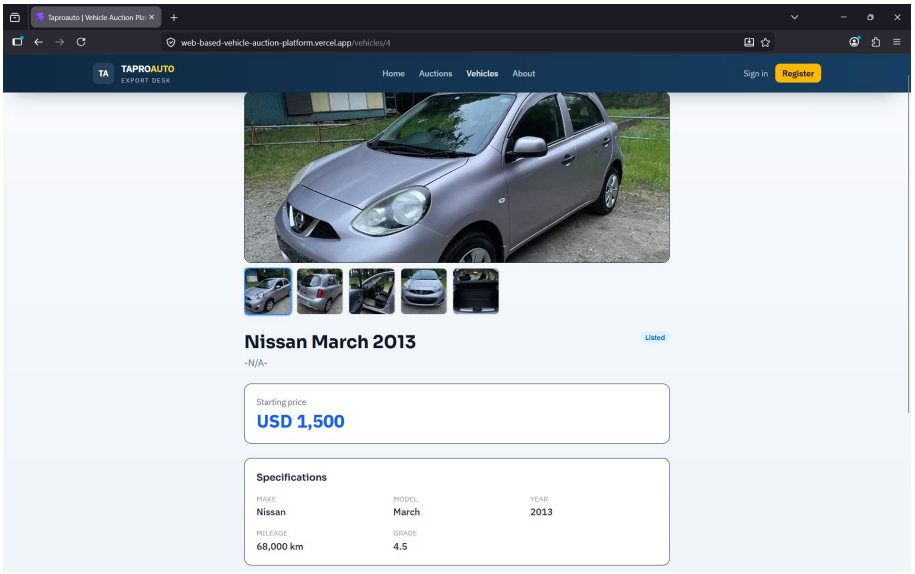


Figure 5.3 A test done by the client

The client provided positive feedback on the core workflow, particularly the transparent public bid history and the admin dashboard's pending-verification badge for tracking new buyer registrations. The main suggestion was to display the reserve status directly in the admin auction list instead of only on the auction details page, leading to a minor UI refinement. No functional defects were identified during UAT; the feedback focused solely on usability improvements, indicating that major issues had already been resolved during earlier testing.

Chapter 6 - Conclusion

6.1 Critical Evaluation

The delivered system meets its core functional requirements: secure registration and authentication, admin-controlled buyer verification, full vehicle CRUD with search and filtering, time-bound auctions with automatically and reliably correct status, minimum-increment bid validation, public bid transparency, hidden reserve pricing with a buyer-visible indicator, and both automatic and manual winner selection. The non-functional requirements around security (parameterised queries, bcrypt hashing, role-based access control enforced server-side) and maintainability (the strict layered backend architecture) were also achieved and were directly responsible for making the defects described in Chapter 5 findable and fixable in isolation , for example, the bid-scoping defect (Bug #5) required changes only to the bids repository and a migration, not to any controller or route.

Some trade-offs were made deliberately. The JWT-based authentication scheme accepts a known staleness window: a buyer's verification status embedded in their token does not update until they log in again, even if an admin verifies them in the interim. This was judged an acceptable cost for the simplicity and statelessness gained by avoiding a server-side session store, and is mitigated by having the profile page always read live data from /api/users/me rather than trusting the token. Similarly, the dual-layer auction status correction (cron plus on-read) accepts up to a one-minute window where the database's stored status could theoretically lag reality if no request happens to arrive in that window; in practice, the on-read correction means no client-visible response is ever stale, only the raw database row briefly is.

The system's main current limitation, by deliberate scope decision rather than oversight, is the absence of payment and shipping-logistics integration, meaning the platform manages the auction and winner-selection process but the actual transaction and delivery still proceed offline, as documented in Section 1.4.

6.2 Personal Reflection

This project required applying software development principles across the full stack rather than in isolation, and the most valuable learning came specifically from the defect-discovery process documented in Chapter 5. Several of the defects found - particularly the bid-scoping architectural flaw and the login-blocking authentication bug - were not surface-level mistakes but consequences of an initial design that had not been thought through against the actual business requirement closely enough; finding and correctly diagnosing them required systematically tracing requirements through to implementation rather than only testing happy-path behaviour. Working independently, without a team to catch design flaws through review, reinforced the value of writing the layered architecture early, since it made these defects locatable and fixable in single, contained files rather than requiring wide, risky refactors. The experience also developed practical skills in incremental database schema management through migrations, in writing tests that catch real defects (Bug #11 was found by a test assertion, not by manual inspection), and in communicating technical trade-offs (such as the JWT staleness window) to a non-technical client in a way that let them make an informed decision rather than treating it as a hidden flaw.

6.3 Future Work

Several extensions were identified jointly with the client as valuable next steps beyond the current scope. Payment integration, allowing a winning buyer to place a deposit or full payment directly through the platform, was the most frequently requested addition during client discussions, though it was correctly excluded from this project's scope given the additional compliance and security surface area it would introduce. Shipping and logistics tracking, allowing the client to record and the buyer to view the export and delivery status of a won vehicle, would extend the system's usefulness beyond the point of sale. Real-time bid updates via WebSockets (Socket.io) could replace the current 15-second polling interval on the auction detail page, reducing latency between a bid being placed and other viewers seeing it, at the cost of additional infrastructure complexity that was judged unnecessary at the project's current scale. Finally, basic analytics , auction-level summaries of bid counts, participation rates, and price trends over time , would help the client identify which categories of vehicle attract the strongest buyer interest, supporting better purchasing decisions at the Japanese auctions that supply the platform.

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Appendices

8.1 Appendix A - System Manual

Web-Based Vehicle Auction Platform

System Manual

Appendix A , Installation, Compilation, and Execution Guide

Stack: Node.js · Express · PostgreSQL · React 19 · Vite · Tailwind CSS v4

This manual documents how to install, configure, run, and extend the submitted codebase. It does not describe building the project from scratch , the code already exists in Backend/ and Frontend/.

1. Repository Layout

```
Backend/
  src/{config,db,middleware,repositories,services,controllers,routes}
  migrations/        (001-009, numbered SQL files)
  scripts/seed-admins.js
Frontend/
  src/{api,components,context,pages,router}
```

2. Prerequisites

Tool	Version	Source
Node.js	20 LTS	https://nodejs.org
PostgreSQL	16	https://www.postgresql.org
Git	any recent	https://git-scm.com

3. Installation

git clone <repository-url> vehicle-auction
cd vehicle-auction/Backend && npm install
cd ../Frontend && npm install

Table 8.1 Backend dependencies (reusable code, all from the public npm registry)

Package	Purpose	Location
express	HTTP server/routing	https://npmjs.com/package/express
pg	PostgreSQL client/pool	https://npmjs.com/package/pg
bcrypt	Password hashing	https://npmjs.com/package/bcrypt
jsonwebtoken	JWT auth	https://npmjs.com/package/jsonwebtoken
joi	Request validation	https://npmjs.com/package/joi
morgan	HTTP logging	https://npmjs.com/package/morgan
node-cron	Auction status scheduler	https://npmjs.com/package/node-cron

Package	Purpose	Location
dotenv	Env var loading	https://npmjs.com/package/dotenv
nodemon (dev)	Auto-restart	https://npmjs.com/package/nodemon
Nodemailer	Sends password reset emails	https://npmjs.com/package/nodemailer

Table 8.2 Frontend dependencies

Package	Purpose	Location
react, react-dom	UI framework	https://npmjs.com/package/react
react-router-dom	Client routing	https://npmjs.com/package/react-router-dom
axios	HTTP client	https://npmjs.com/package/axios
tailwindcss, @tailwindcss/vite, @tailwindcss/forms	Styling	https://tailwindcss.com
vite, @vitejs/plugin-react	Build tool	https://vitejs.dev

External service (not npm-hosted)

Cloudinary (image CDN, free tier) (cloudinary.com)used for unsigned direct browser-to-CDN image upload. No image binaries are ever stored in the app or database.

4. Environment Variables

Backend/.env (never committed):

```
PORT=3000
DATABASE_URL=postgres://user:password@localhost:5432/vehicle_auction
JWT_SECRET=replace-me
JWT_EXPIRES_IN=1h
BCRYPT_ROUNDS=10
PASSWORD_RESET_EXPIRES_MINUTES=60
APP_BASE_URL=http://localhost:5173
SMTP_HOST=
SMTP_PORT=587
SMTP_SECURE=false
SMTP_USER=
SMTP_PASS=
EMAIL_FROM=no-reply@vehicle-auction.local
NODE_ENV=development
```

Frontend/.env:

```
VITE_CLOUDINARY_CLOUD_NAME=<your cloud name>
VITE_CLOUDINARY_UPLOAD_PRESET=<your unsigned preset>
```

5. Database Setup

```
createdb vehicle_auction
for f in Backend/migrations/*.sql; do psql -d vehicle_auction -
f "$f"; done
node Backend/scripts/seed-admins.js    # creates first admin
account
```

Migrations are numbered 001-009 and must be applied in order; never edit an applied migration , add a new one instead. See Chapter 3 (Design) of the main dissertation for the full rationale behind each migration.

6. Running the Application

```
npm run dev      # development, auto-restart
npm start        # production

# Frontend (from Frontend/)
npm run dev         # development server, proxies /api to
localhost:3000
npm run build    # production build → Frontend/dist
```

Health check: GET http://localhost:3000/api/health → 200 OK.

7. Compilation Notes

- Backend is plain ECMAScript-module JavaScript , no compile step; "type": "module" must remain in Backend/package.json.
- Frontend is compiled/bundled by Vite via npm run build, producing static assets in Frontend/dist/.

8. API Route Map

Method	Path	Auth	Description
POST	/api/auth/register	None	Register buyer
POST	/api/auth/login	None	Login
GET	/api/users/me	Auth	Own profile
PATCH	/api/users/:id/status	Admin	Verify/reject buyer
GET/POST/PUT/DELETE	/api/vehicles[/:id]	Mixed	Catalogue CRUD
GET/POST/PUT/DELETE	/api/auctions[/:id]	Mixed	Auction CRUD
GET	/api/auctions/won/me	Auth	Must be declared before /:id

Method	Path	Auth	Description
POST	/api/auctions/:id/close	Admin	Close + auto-assign winner
POST	/api/auctions/:id/winner	Admin	Manual winner override
GET/POST	/api/auctions/:id/bids	Public/Verified buyer	Bid history / place bid
POST	/api/auth/forgot-password	None	Request password reset email
POST	/api/auth/reset-password	None	Consume token, set new password

9. Actual Deployment (as delivered)

The delivered system is hosted on:

- Backend + PostgreSQL: Railway (managed hosting, automated build from Git)
- Frontend: Vercel (static hosting + CDN)

Steps:

- Push Backend/ and Frontend/ to Git.
- On Railway: create a Postgres instance, deploy Backend/ as a service, set the environment variables from §4, run the migrations and seed script via Railway's shell.
- On Vercel: import Frontend/, set the VITE_* env vars, deploy. Vercel serves the SPA with automatic fallback routing (no manual Nginx try_files config needed).
- Point the frontend's API base URL at the Railway backend's public URL.

A self-hosted Ubuntu/Nginx/PM2 deployment is also technically possible with this codebase, but is not how the delivered system is run, and is not documented here to avoid conflicting with the actual production setup.

10. Known Limitations for Future Extension

Anyone extending this codebase should be aware the following requirements were descoped and are not implemented:

- Buyer registration does not currently capture a contact/phone number (Joi schema and DB insert only cover name/email/password).
- No email or in-app notification system (verification status, outbid alerts) exists.
- No batch/CSV vehicle upload.
- Payment processing and shipping/logistics tracking are out of scope.

8.2 Appendix B - User Manual

User Manual

Web-Based Vehicle Auction Platform

1. Introduction

This manual explains how to use the Web-Based Vehicle Auction Platform. It is written for three types of readers: Guests browsing the public catalogue, registered Buyers who place bids, and Administrators who manage vehicles, auctions, and user accounts. Each section below is organised around a specific task, and screenshots are provided to show exactly what you will see on screen.

1.1 Who This Manual Is For

Reader	Relevant Sections
Guest (not logged in)	Sections 3, 4.1 - 4.5
Registered Buyer	Sections 3, 4.1 - 4.9, 6
Administrator	Sections 3, 5.1 - 5.5, 6

1.2 Accessing the System

The platform is accessed through any modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari). No installation is required. Enter the system's web address in the browser's address bar to open the vehicle catalogue.

Recommended screen resolution: 1280 × 720 or higher for the best layout on the admin dashboard. The site is also usable on mobile phones and tablets.

2. User Roles Overview

The system defines three levels of access. A visitor's capabilities expand as they register and are verified by an administrator.

Role	How Obtained	Capabilities
Guest	No action needed , default for any visitor	Browse the vehicle catalogue and auctions; view bid history and winner pages
Buyer (unverified)	Self-registration at the Register page	All Guest capabilities; cannot place bids until verified by an administrator
Buyer (verified)	Approved by an administrator after registration	All Guest capabilities plus placing bids on active auctions

Role	How Obtained	Capabilities
Administrator	Created only by the system owner (not self-service)	Full control: manage vehicles, auctions, and users; verify buyers; close auctions and select winners

3. Getting Started

3.1 Creating an Account

New buyers must register before they can bid. Follow these steps:

1. Open the platform in your browser and click Register in the top navigation bar.
2. Fill in your full name, email address, and a password.
3. Click Create Account.
4. You will be redirected to the Login page with a confirmation message that your account was created.

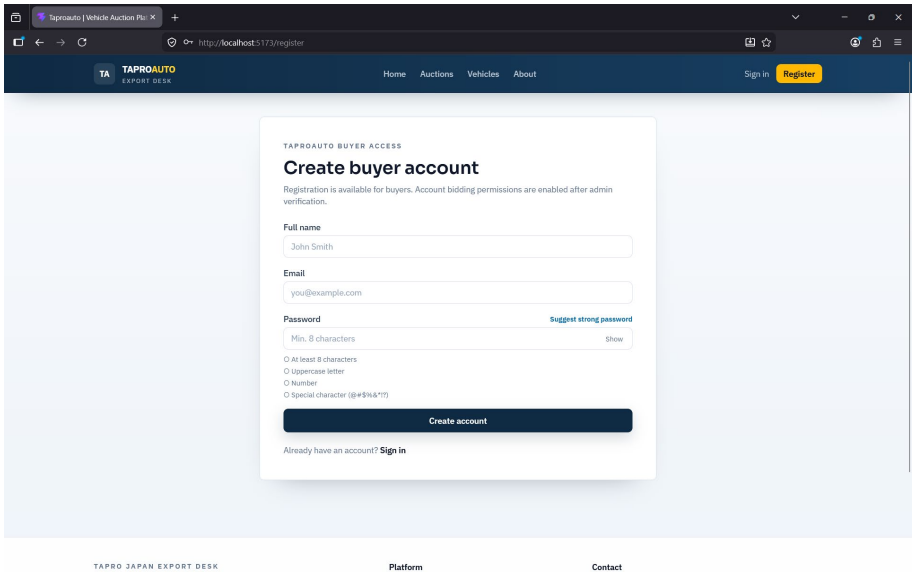


Figure 8.1 Create a Buyer Account

Note: A newly registered account cannot place bids immediately. An administrator must verify your account first - see Section 4.6.

3.2 Logging In

5. Click Login in the top navigation bar.
6. Enter your registered email address and password.
7. Click Login.

On successful login you are redirected automatically: administrators are taken to the Admin Dashboard, and buyers are taken to the Auctions page.

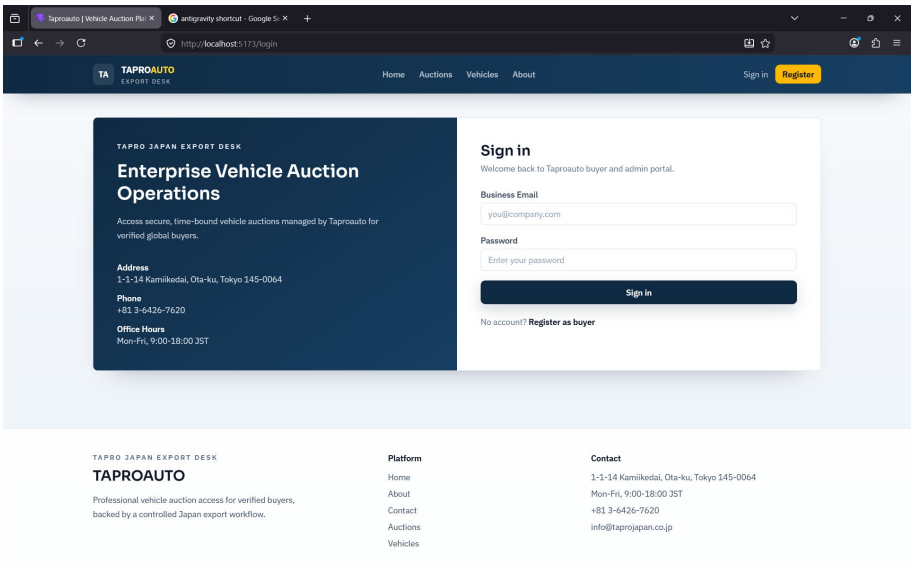


Figure 8.2 Logging In

3.3 Navigating the Homepage

The top navigation bar is visible on every page and provides links to Vehicles, Auctions, and (once logged in) your Profile. Administrators additionally see an Admin link with a red badge showing the number of buyer accounts awaiting verification.

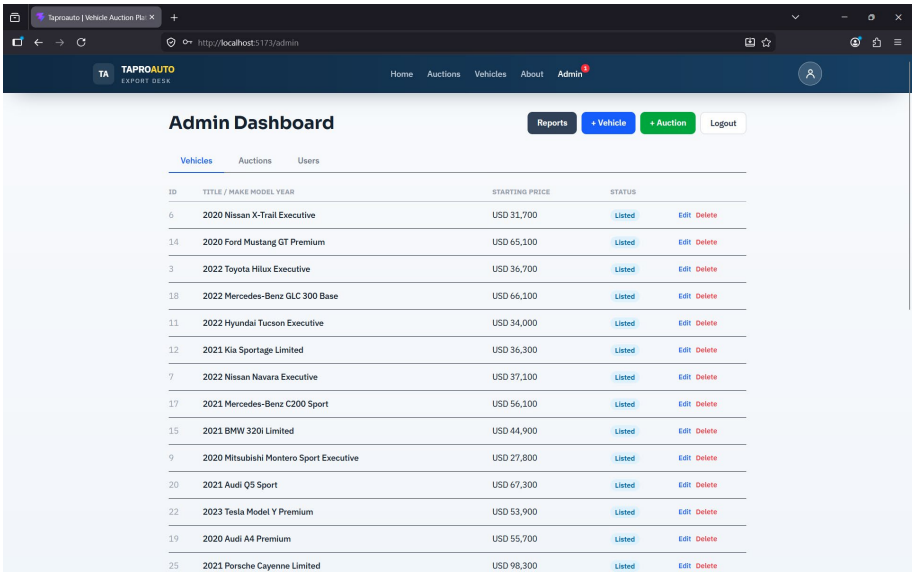


Figure 8.3 Admin link with a red badge showing the number of buyer accounts awaiting verification

4. Buyer Features

4.1 Browsing the Vehicle Catalogue

The Vehicles page lists all listed vehicles as cards showing a thumbnail image, make, model, year, and starting price. Use the pagination controls at the bottom of the page to move between pages of results.

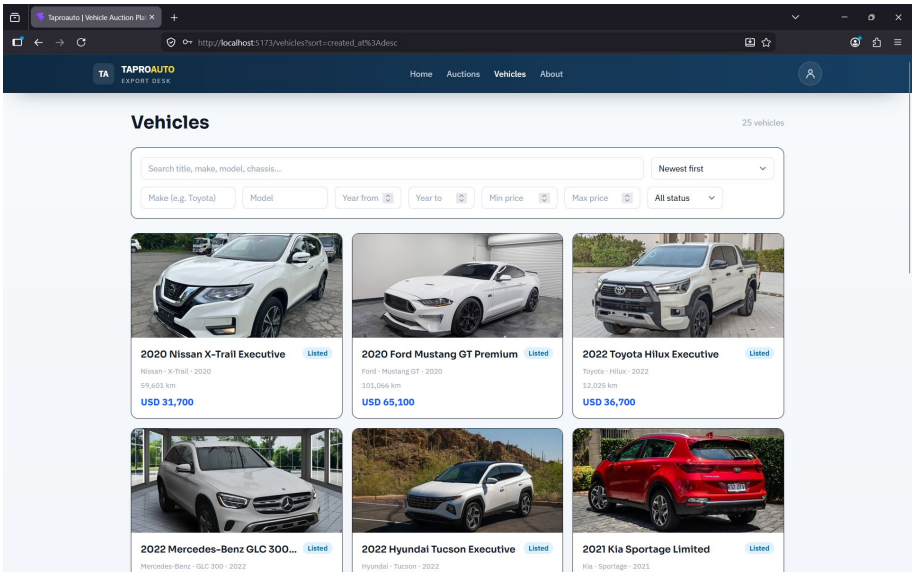


Figure 8.4 Vehicles Page

4.2 Filtering and Searching Vehicles

Use the filter panel at the top of the Vehicles page to narrow results by make, model, year range, and price range. Filters are reflected in the page's web address, so you can bookmark or share a filtered search with someone else.

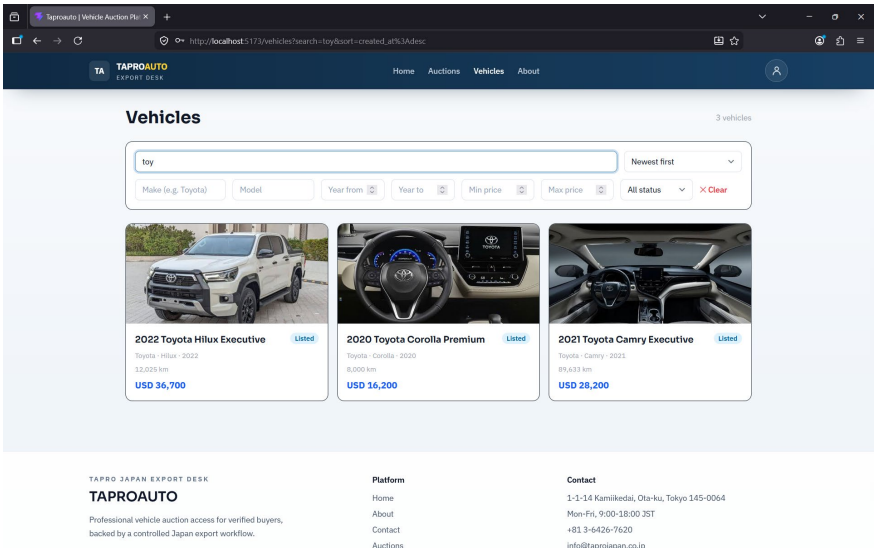


Figure 8.5 Vehicle Filtering and Searching

4.3 Viewing Vehicle Details

Click any vehicle card to open its detail page, which shows the full specification (chassis number, mileage, grade), an image gallery, and a link to its associated auction if one exists.

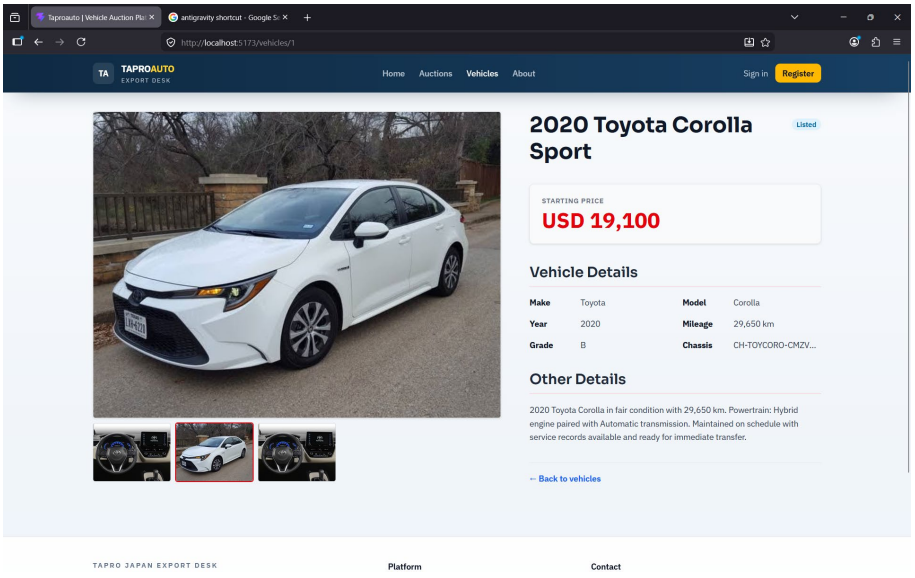


Figure 8.6 Viewing Vehicle Details

4.4 Browsing Auctions

The Auctions page lists all auctions with a status badge (Draft, Active, or Ended) and, for active auctions, a live countdown timer showing the time remaining.

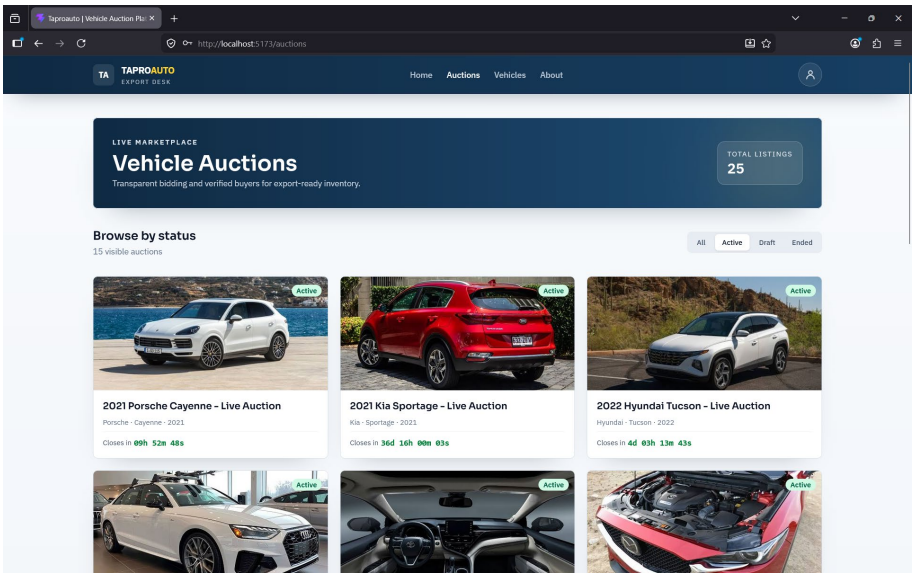


Figure 8.7 Vehicle Auctions

4.5 Viewing Auction Details

Opening an auction shows the vehicle's image gallery (click the main image for a full-screen view; use the arrow keys to move between images and Escape to close), the current highest bid, the full bid history, and a reserve price indicator.

The reserve indicator shows one of three states: no reserve set, reserve not yet met, or reserve met. Buyers see only this indicator; the actual reserve amount is visible to administrators only.

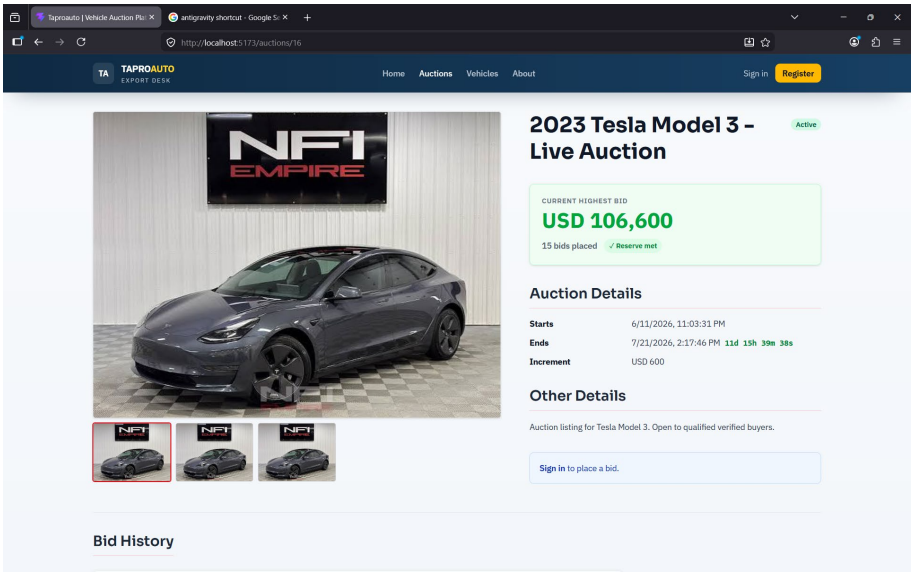


Figure 8.8 Vehicle Auction Details

Note: The auction page automatically refreshes bid data every 15 seconds while the auction is active, so you do not need to reload the page to see new bids.

4.6 Account Verification

Before you can place a bid, an administrator must verify your account (see Section 5.4). You can check your verification status at any time on your Profile page. If you log in before your account is verified, log out and log back in after verification so your session reflects the update.

4.7 Placing a Bid

Once verified, a bid form appears at the bottom of any active auction's detail page.

8. Open the detail page of an active auction.
9. Enter a bid amount that is at least the current highest bid plus the minimum increment shown on the page.
10. Click Place Bid.
11. Your bid appears immediately at the top of the bid history table.

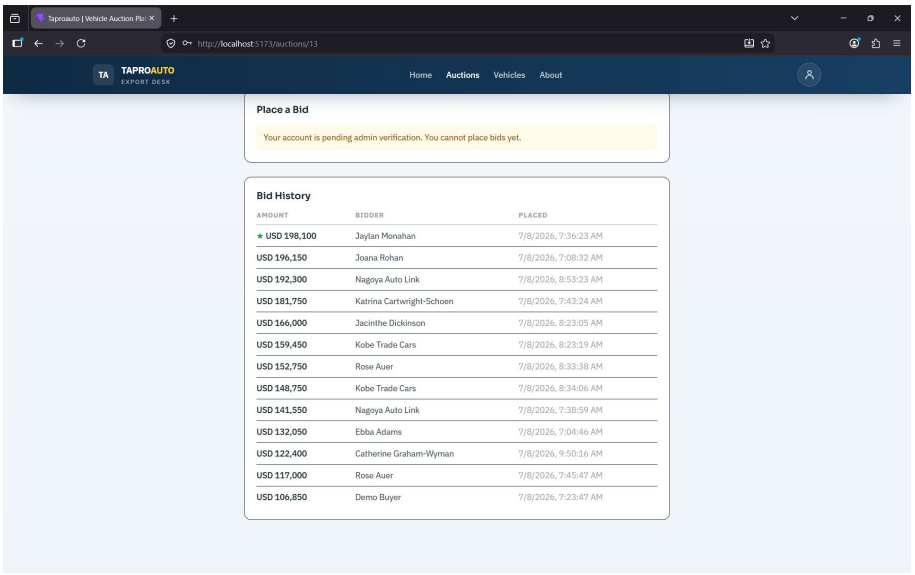


Figure 8.9 Bid History View

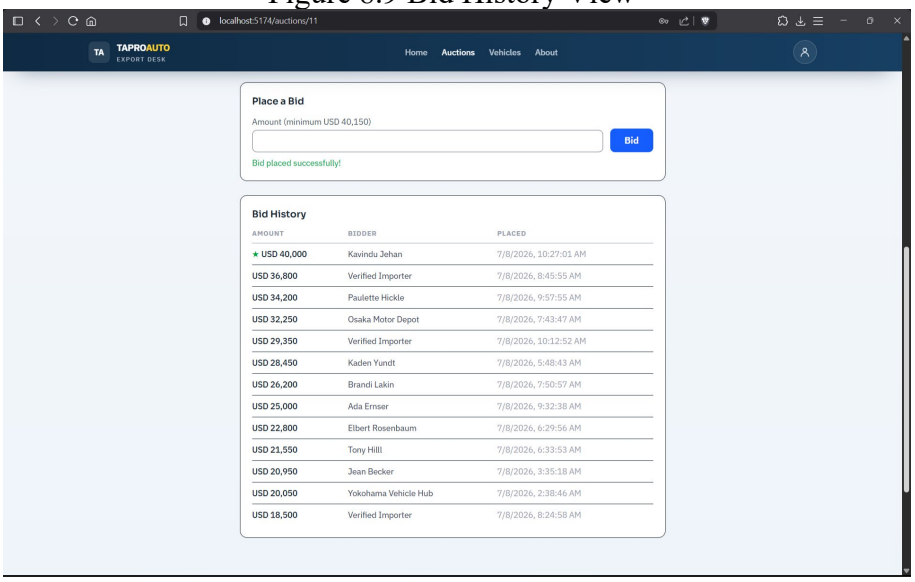


Figure 8.10 Placing a Bid

Note: If your bid is below the required minimum, the system rejects it and displays an error message stating the minimum acceptable amount.

4.8 Winning an Auction

When an auction closes, the highest bid is either automatically selected as the winner or confirmed by an administrator. If you win, a green "You Won!" banner appears at the top of the site the next time you visit while logged in. Click the banner's close icon to dismiss it , it will not reappear.

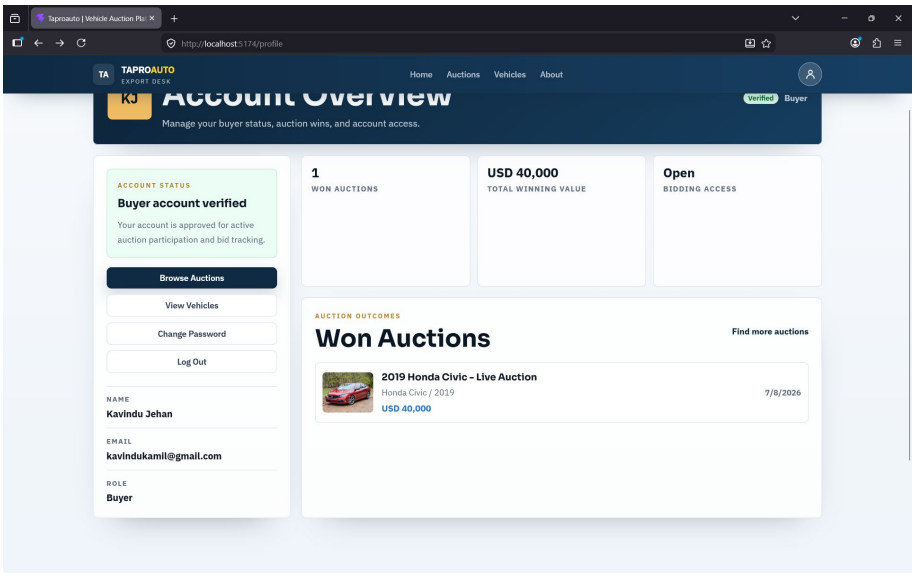


Figure 8.11 Won Auction Details

Anyone can view the public Winner page for a closed auction, which shows the winning buyer's name and the winning bid amount.

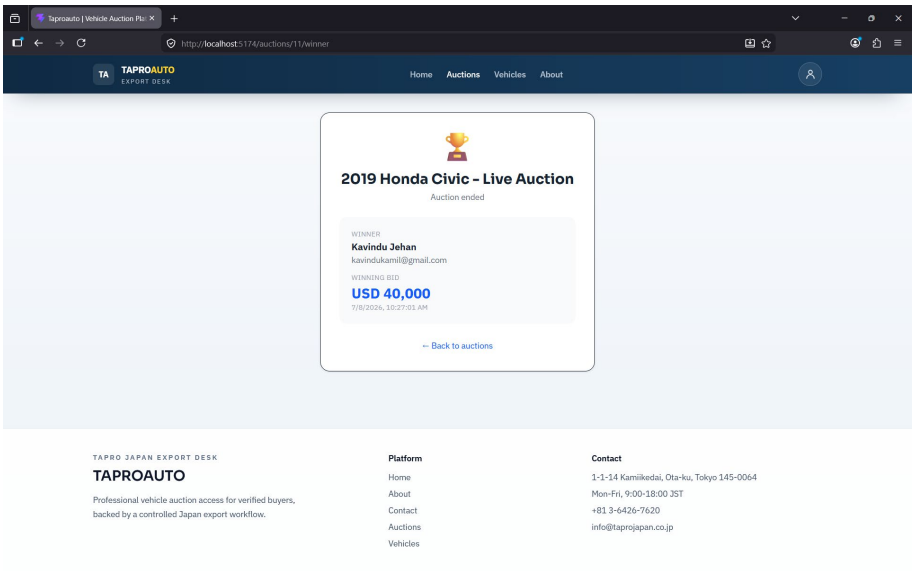


Figure 8.12 Auction Won Page

4.9 My Profile

The Profile page shows your account details, current verification status, and a list of auctions you have won, including the winning bid amount for each.

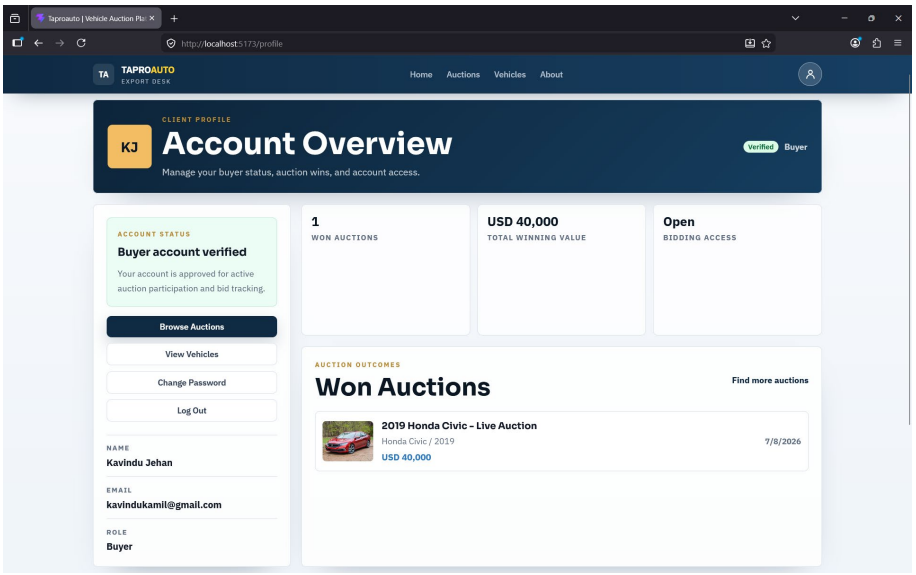


Figure 8.13 User Profile

5. Administrator Features

5.1 Accessing the Admin Dashboard

Log in with an administrator account to be redirected automatically to the Admin Dashboard, available at any time from the Admin link in the navigation bar. The dashboard has three tabs: Vehicles, Auctions, and Users.

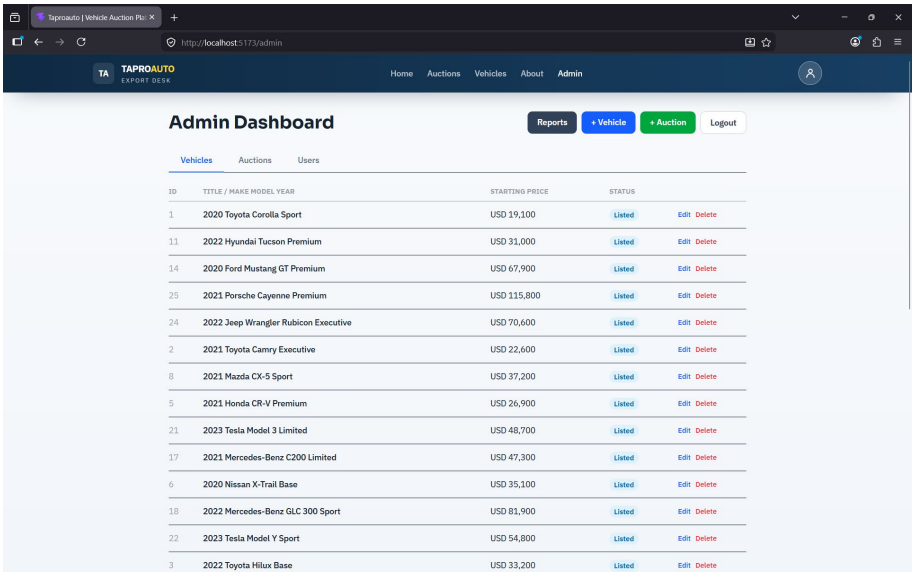


Figure 8.14 Admin Dashboard

5.2 Managing Vehicles

The Vehicles tab lists every vehicle with options to edit or delete each entry, and an Add Vehicle button to create a new listing.

12. Click Add Vehicle (or the edit icon on an existing vehicle).
13. Fill in the title, description, make, model, year, starting price, chassis number, mileage, and grade.
14. Upload one or more images , each is uploaded to the image hosting service immediately and a thumbnail preview appears.
15. Click Save. The vehicle now appears in the public catalogue if its status is Listed.

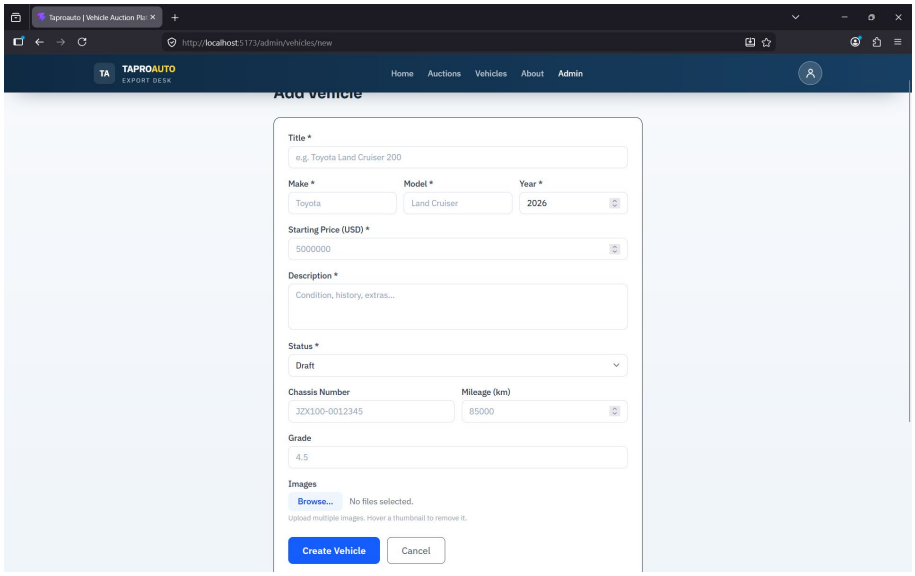


Figure 8.15 Add Vehicle Page

5.3 Managing Auctions

The Auctions tab lists all auctions. Use Add Auction to link a new auction to an existing vehicle, setting a start time, end time, minimum bid increment, and an optional reserve price. Administrators alone can see the actual reserve price amount on the auction detail page.

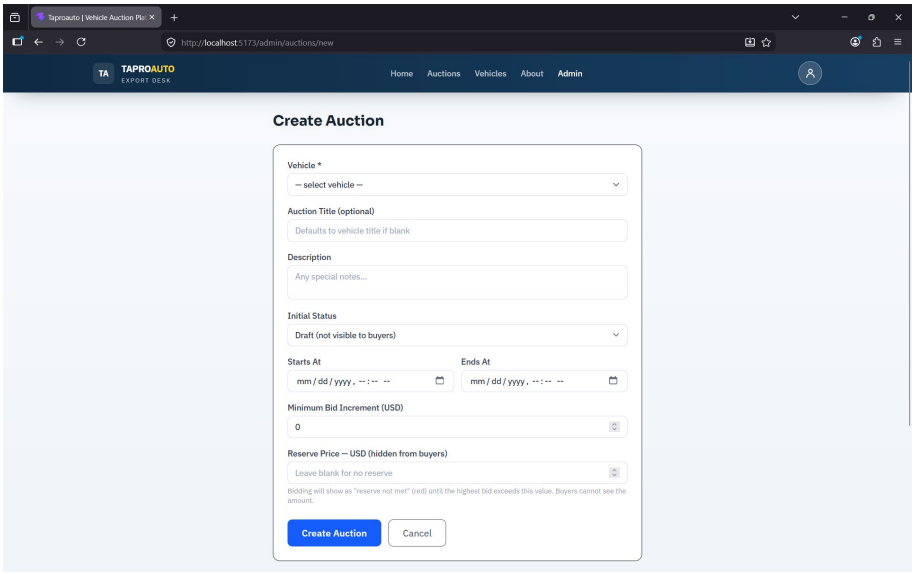


Figure 8.16 Auctions Page

Two ways exist to end an auction:

- Automatically - the system checks every minute and moves an auction from Active to Ended once its end time has passed.
- Manually - click Close Auction on the auction detail page; this immediately ends the auction and assigns the current highest bid as the winner.

After closing, review the bid ladder and use Select Winner if you need to override the automatically assigned winner (for example, if the highest bid did not meet the reserve price).

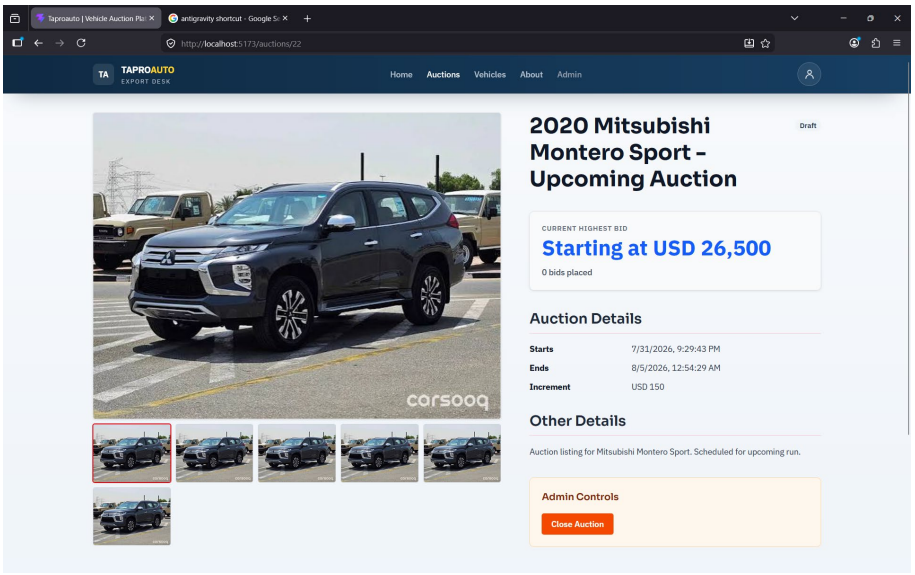


Figure 8.17 Managing Auctions

5.4 Managing Users

The Users tab lists every registered buyer with their current verification status: Pending, Verified, or Rejected.

16. Locate the buyer awaiting verification (shown as Pending).
17. Click Verify to approve the account, or Reject to deny it.
18. The pending-count badge on the Admin navigation link updates immediately without a page reload.

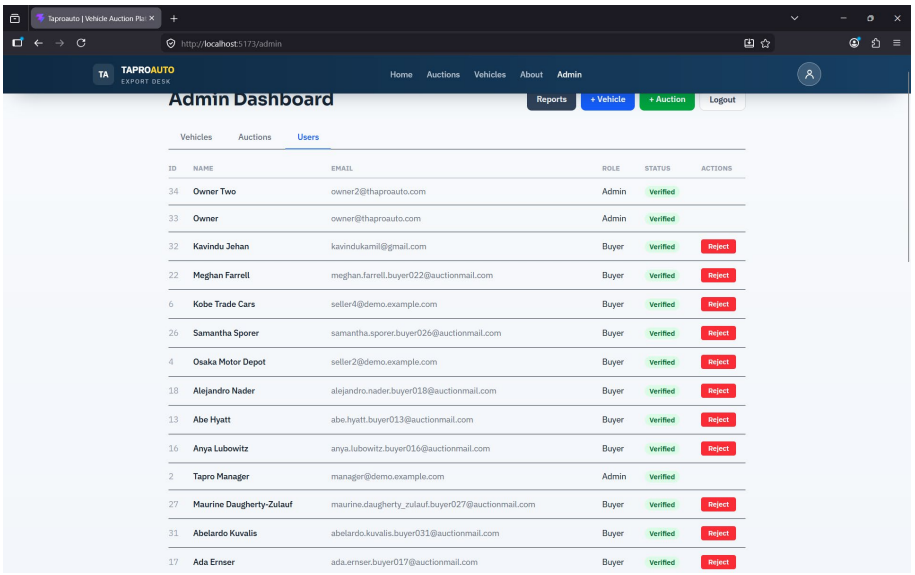


Figure 8.18 Managing Users

Note: A rejected buyer can be re-verified later if circumstances change; rejection is not permanent.

5.5 Reserve Price Visibility

Only administrators can see the exact reserve price figure on an auction. Buyers instead see a colour-coded indicator (no reserve, reserve not met, or reserve met) so pricing strategy stays confidential while bidders still know where they stand.

6. Account Security

6.1 Automatic Logout

For your security, you are automatically logged out after 30 minutes of inactivity. Any mouse movement, click, keystroke, or scroll resets this timer. If you are logged out, you will see a session-expired message on the Login page and can simply log in again.

6.2 Changing Your Password

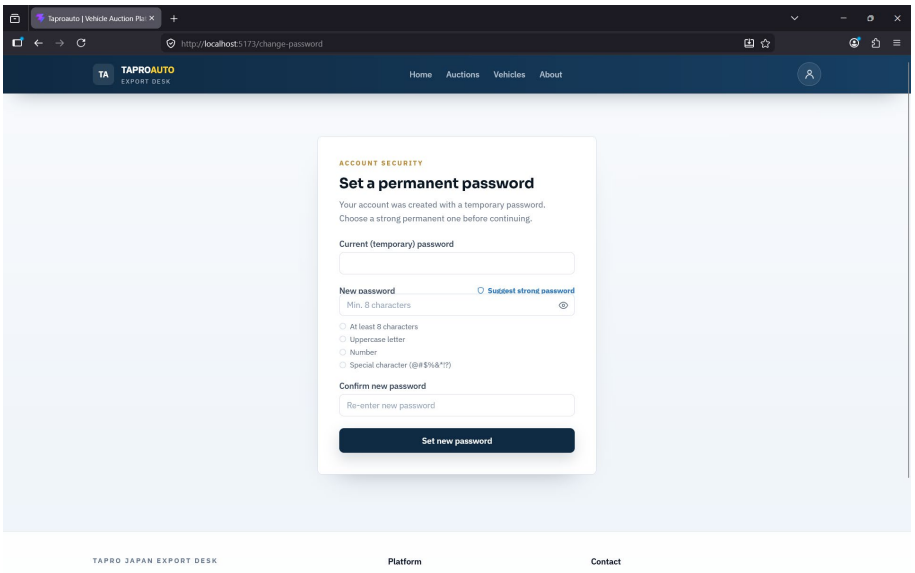


Figure 8.19 Change Password

If you have forgotten your password, click **Forgot Password?** on the Login page and enter the email address associated with your account. The system will send a password reset link to your registered email address. Open the email, click the link, and enter your new password to complete the reset process. Password reset links are valid only for a limited period and can be used only once. If the link expires or no email is received, request a new reset link and check your spam or junk folder before contacting an administrator.

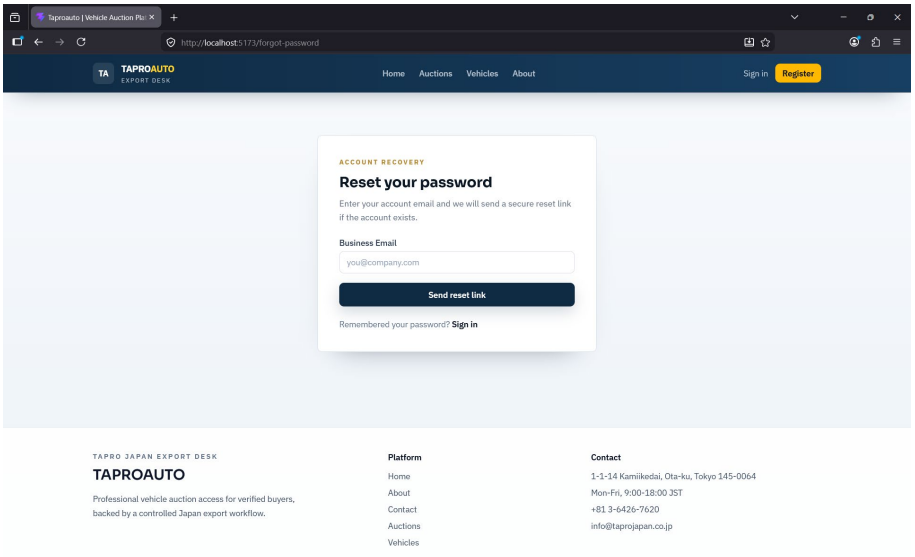


Figure 8.20 Reset Password

7. Troubleshooting

Issue	What To Do
I cannot see the bid form on an active auction.	Confirm you are logged in and that your account has been verified by an administrator. Log out and log back in after verification.
My bid was rejected.	Your bid amount was below the current highest bid plus the minimum increment. Check the amounts shown on the auction page and try a higher bid.
I was logged out unexpectedly.	Sessions expire after 30 minutes of inactivity for security. Simply log in again.
Images are not uploading on the vehicle form.	Check your internet connection and that the file is a supported image format (JPG or PNG). Try a smaller file if the upload times out.
The pending-users badge did not update.	Refresh the page. The badge updates automatically after a Verify or Reject action but requires the browser tab to remain open.
Password Reset Email Not Received	Verify that the correct email address was entered.Check the spam or junk mail folder.

8. Support

For further assistance beyond this manual, contact the system administrator or the development team using the contact details provided separately by TaproJapan Co. Ltd.

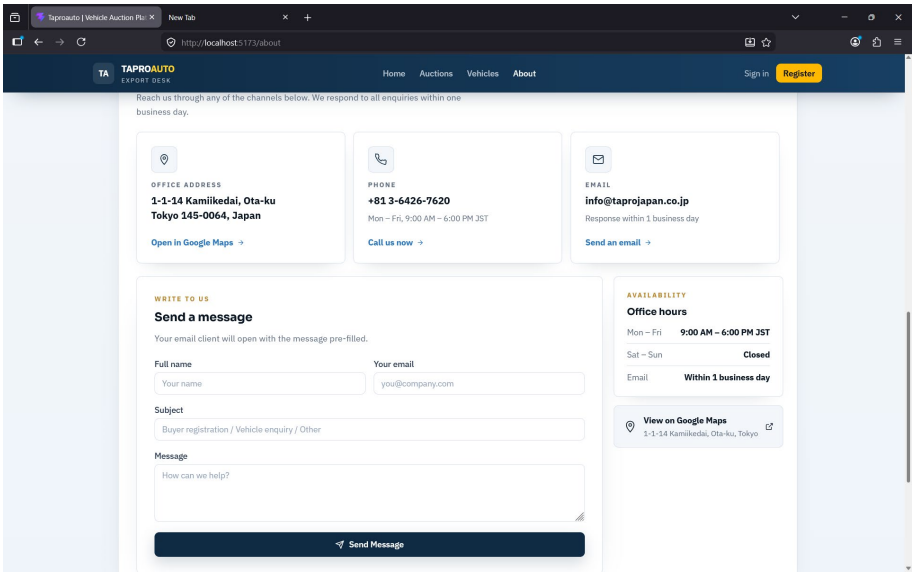
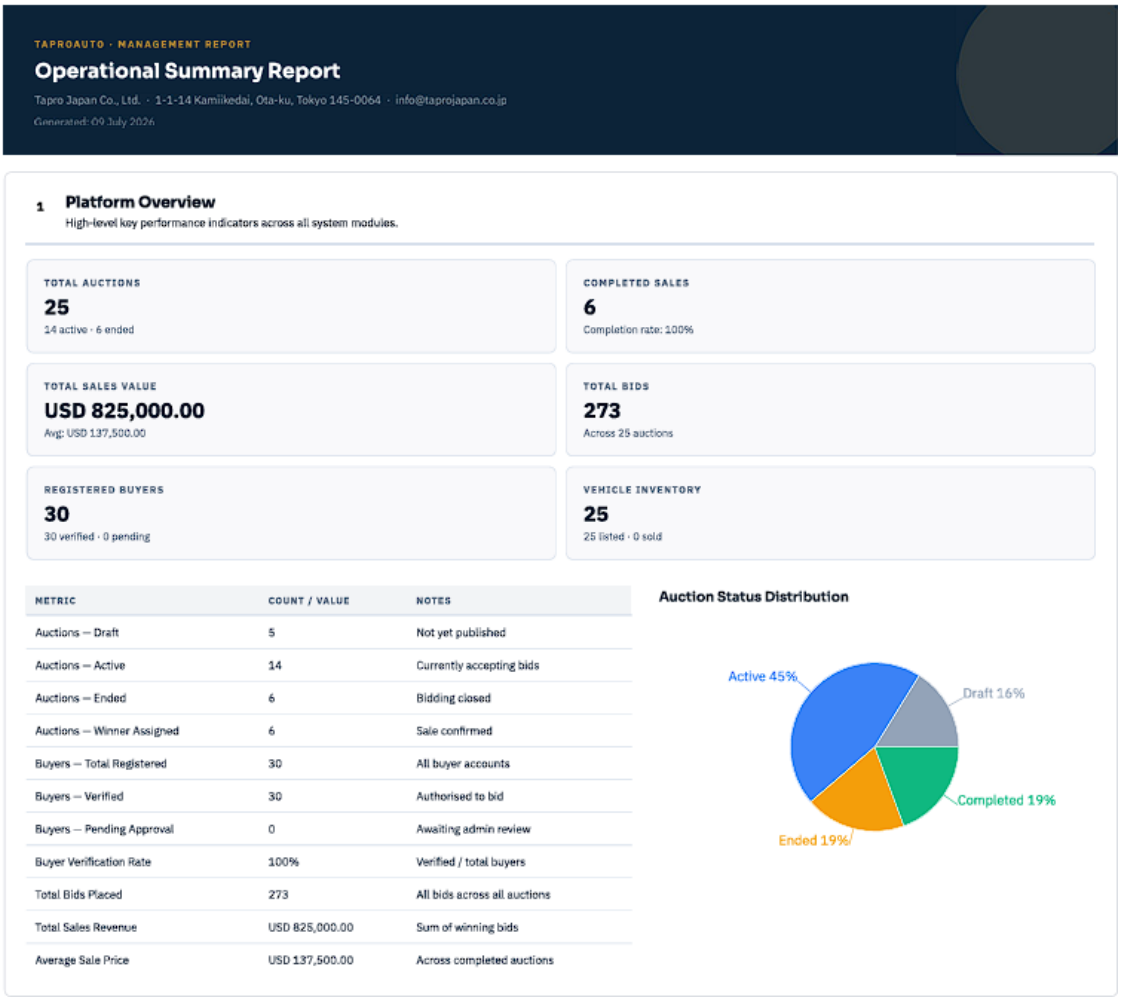


Figure 8.21 Contact Details

Appendix C - Management Report



TAPROAUTO · MANAGEMENT REPORT

Operational Summary Report

Tapro Japan Co., Ltd. · 1-1-14 Kamiikedai, Ota-ku, Tokyo 145-0064 · info@taprojapan.co.jp

Generated: 09 July 2026

1 Platform Overview

High-level key performance indicators across all system modules.

TOTAL AUCTIONS **25** 14 active · 6 ended	COMPLETED SALES **6** Completion rate: 100%
TOTAL SALES VALUE **USD 825,000.00** Avg: USD 137,500.00	TOTAL BIDS **273** Across 25 auctions
REGISTERED BUYERS **30** 30 verified · 0 pending	VEHICLE INVENTORY **25** 25 listed · 0 sold

METRIC	COUNT / VALUE	NOTES
Auctions — Draft	5	Not yet published
Auctions — Active	14	Currently accepting bids
Auctions — Ended	6	Bidding closed
Auctions — Winner Assigned	6	Sale confirmed
Buyers — Total Registered	30	All buyer accounts
Buyers — Verified	30	Authorised to bid
Buyers — Pending Approval	0	Awaiting admin review
Buyer Verification Rate	100%	Verified / total buyers
Total Bids Placed	273	All bids across all auctions
Total Sales Revenue	USD 825,000.00	Sum of winning bids
Average Sale Price	USD 137,500.00	Across completed auctions

Auction Status Distribution

Figure 8.22 Summary Report Page 1

2

Auction Performance Report

Detailed breakdown of all 25 auction(s) conducted on the platform.

Winning Amount vs Starting Price



#	AUCTION TITLE	VEHICLE	PERIOD	STATUS	STARTING PRICE	HIGHEST BID	WINNING BID	WINNER	BIDS	RESERVE
1	2020 Mitsubishi Montero Sport - Upcoming Auction	Mitsubishi Montero Sport 2020	31 Jul 2026 05 Aug 2026	Draft	USD 26,500.00	–	–	–	0	–
2	2021 Suzuki Swift - Upcoming Auction	Suzuki Swift 2021	30 Jul 2026 02 Aug 2026	Draft	USD 16,000.00	–	–	–	0	–
3	2021 Toyota Camry - Upcoming Auction	Toyota Camry 2021	20 Jul 2026 26 Jul 2026	Draft	USD 22,600.00	–	–	–	0	–
4	2022 Toyota Hilux - Upcoming Auction	Toyota Hilux 2022	16 Jul 2026 20 Jul 2026	Draft	USD 33,200.00	–	–	–	0	Not Met
5	2022 BMW X5 - Upcoming Auction	BMW X5 2022	14 Jul 2026 19 Jul 2026	Draft	USD 88,700.00	–	–	–	0	Not Met
6	2019 Honda Civic - Live Auction	Honda Civic 2019	08 Jul 2026 08 Jul 2026	Ended	USD 17,700.00	USD 40,000.00	USD 40,000.00	Kavindu Jehan	13	–
7	2021 Mazda CX-5 - Live Auction	Mazda CX-5 2021	05 Jul 2026 13 Aug 2026	Active	USD 37,200.00	USD 55,300.00	–	–	9	Met
8	2022 Hyundai Tucson - Live Auction	Hyundai Tucson 2022	03 Jul 2026 10 Jul 2026	Active	USD 31,000.00	USD 35,900.00	–	–	4	Not Met
9	2020 Nissan X-Trail - Live Auction	Nissan X-Trail 2020	03 Jul 2026 14 Aug 2026	Active	USD 35,100.00	USD 54,950.00	–	–	9	–
10	2021 Kia Sportage - Live Auction	Kia Sportage 2021	03 Jul 2026 05 Aug 2026	Active	USD 31,300.00	USD 56,350.00	–	–	13	Met
11	2021 Honda CR-V - Live Auction	Honda CR-V 2021	24 Jun 2026 04 Aug 2026	Active	USD 26,900.00	USD 45,700.00	–	–	11	Met
12	2021 Audi Q5 - Live Auction	Audi Q5 2021	24 Jun 2026 22 Jul 2026	Active	USD 75,800.00	USD 203,700.00	–	–	16	Met
13	2021 Mercedes-Benz C200 - Live Auction	Mercedes-Benz C200 2021	23 Jun 2026 13 Aug 2026	Active	USD 47,300.00	USD 124,100.00	–	–	17	–
Totals					USD 1,175,400.00		USD 825,000.00		273	

Figure 8.23 Auction Performance Report



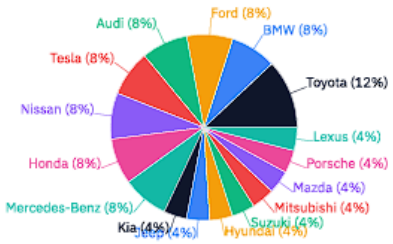
#	BUYER NAME	EMAIL	REGISTERED	STATUS	BIDS PLACED	AUCTIONS ENTERED	AUCTIOI
1	Orlando Conn	orlando.conn.buyer012@auctionmail.com	22 Dec 2025	Verified	15	11	
2	Paulette Hickle	paulette.hickle.buyer029@auctionmail.com	15 Jan 2026	Verified	13	9	
3	Osaka Motor Depot	seller2@demo.example.com	24 May 2026	Verified	13	11	
4	Ada Ernser	ada.ernser.buyer017@auctionmail.com	29 Mar 2026	Verified	13	10	
5	Meghan Farrell	meghan.farrell.buyer022@auctionmail.com	23 Jun 2026	Verified	12	9	
6	Izabella Berge	izabella.berge.buyer024@auctionmail.com	10 Mar 2026	Verified	12	7	
7	Mattie Hegmann	mattie.hegmann.buyer011@auctionmail.com	04 Jan 2026	Verified	12	9	
8	Demo Buyer	buyer@demo.example.com	24 Jan 2026	Verified	11	9	
9	Brandi Lakin	brandi.lakin.buyer030@auctionmail.com	07 Dec 2025	Verified	11	10	
10	Tony Hilll	tony.hilll.buyer021@auctionmail.com	18 Nov 2025	Verified	10	8	
11	Verified Importer	importer@demo.example.com	07 Feb 2026	Verified	10	8	
12	Kobe Trade Cars	seller4@demo.example.com	09 Jun 2026	Verified	10	9	
13	Nagoya Auto Link	seller3@demo.example.com	19 Mar 2026	Verified	10	7	
14	Anya Lubowitz	anya.lubowitz.buyer016@auctionmail.com	22 Apr 2026	Verified	10	7	

Figure 8.24 Buyer Activity Report

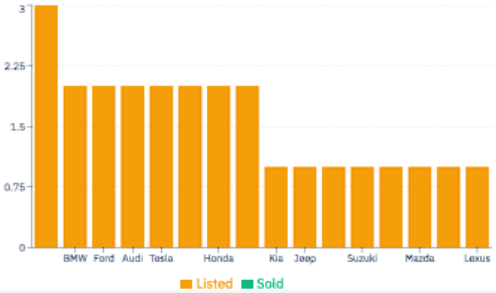
4 **Vehicle Inventory Report**

Breakdown of vehicle stock by make, including status counts and price statistics.

Inventory Distribution by Make



Listed vs Sold by Make



MAKE	TOTAL	DRAFT	LISTED	SOLD	AVG STARTING PRICE	MIN PRICE	MAX PRICE
Toyota	3	0	3	0	USD 24,966.67	USD 19,100.00	USD 33,200.00
BMW	2	0	2	0	USD 65,050.00	USD 41,400.00	USD 88,700.00
Ford	2	0	2	0	USD 60,900.00	USD 53,900.00	USD 67,900.00
Audi	2	0	2	0	USD 57,850.00	USD 39,900.00	USD 75,800.00
Tesla	2	0	2	0	USD 51,750.00	USD 48,700.00	USD 54,800.00
Nissan	2	0	2	0	USD 32,200.00	USD 29,300.00	USD 35,100.00
Honda	2	0	2	0	USD 22,300.00	USD 17,700.00	USD 26,900.00
Mercedes-Benz	2	0	2	0	USD 64,600.00	USD 47,300.00	USD 81,900.00
Kia	1	0	1	0	USD 31,300.00	USD 31,300.00	USD 31,300.00
Jeep	1	0	1	0	USD 70,600.00	USD 70,600.00	USD 70,600.00
Hyundai	1	0	1	0	USD 31,000.00	USD 31,000.00	USD 31,000.00
Suzuki	1	0	1	0	USD 16,000.00	USD 16,000.00	USD 16,000.00
Mitsubishi	1	0	1	0	USD 26,500.00	USD 26,500.00	USD 26,500.00
Mazda	1	0	1	0	USD 37,200.00	USD 37,200.00	USD 37,200.00
Porsche	1	0	1	0	USD 115,800.00	USD 115,800.00	USD 115,800.00
Lexus	1	0	1	0	USD 62,800.00	USD 62,800.00	USD 62,800.00
All Makes	25	0	25	0			

Figure 8.25 Vehicle Inventory Report